

HUMAIN



HUMAIN Create Studio

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Master Deck

PRD · UI/UX · Architecture — Agentic, Bilingual Content and Experience Platform

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Contents

- **Part I — Product Requirements** — vision, scope, requirements, business impact, the agentic platform, and product snapshots.
- **Part II — UI/UX & User Journeys** — design system, navigation, interaction patterns, and the journey for every capability.
- **Part III — High-Level Architecture** — system context, agentic core, data, integration, deployment and security.

Part I — Product Requirements

1. Executive Summary

HUMAIN Create Studio is a **bilingual (Arabic + English), agentic AI-native content and digital-experience platform** designed to let marketing, communications, editorial and brand teams produce on-brand, governed, multilingual content at a fraction of the cost and time of traditional content management systems.

At its core is a team of **specialised AI agents** that plan and carry out the multi-step work of content production — drafting, translating, optimising, brand-checking and assembling publish-ready experiences — operating through a **governed, audited tool layer** so that every agent action is permission-scoped, logged and reversible, with humans approving anything that goes live.

The platform unifies two experiences over a **single content store and a single brand system**:

- **Create Studio** — a light, prompt-first surface where a person describes intent in natural language and the agents draft, translate, optimise and assemble publish-ready content.
- **Structured CMS** — a governed, enterprise-grade content backend providing the schema, roles, editorial workflow, versioning and audit trail demanded by a regulated, brand-sensitive organisation.

Both surfaces are powered by an **agentic AI backbone**: orchestrated agents that reason over the brief, retrieve relevant context (RAG), call governed content tools, and chain steps to a finished deliverable — every action governed by role-based permissions and a complete audit log.

The objective: reduce content production cost and cycle-time by an order of magnitude, while *raising* brand consistency, governance and Arabic-first quality — all hosted in-Kingdom under HUMAIN's control.

Snapshots of the working platform are in **Appendix A**.

2. Business Context and Drivers

Driver	Current-state challenge	Platform response
Arabic-first communication	Most CMS platforms treat Arabic/RTL as an afterthought; translation is manual and inconsistent	Native EN⇌AR with right-to-left rendering and AI-assisted, tone-preserving translation as a first-class capability
Content velocity	Producing a single bilingual page can take days across writers, translators, designers and reviewers	Prompt-to-draft in minutes; AI assists every step from brief to publish
Brand consistency	Brand drift across teams, agencies and channels	A central brand and design-token system that every piece of content inherits and is checked against
Governance and sovereignty	Sensitive content on foreign-hosted SaaS; weak audit trails	In-Kingdom, self-hosted deployment with role-based access, editorial gates and full audit logging
Cost of tooling	Per-seat SaaS licences + separate translation, SEO and DAM tools	One consolidated platform owned by HUMAIN, no per-seat lock-in

3. Vision and Goals

Vision: *Let anyone at HUMAIN turn an idea into a published, on-brand, bilingual digital experience — by describing it, not by operating a tool.*

Primary goals

1. **G1 — Speed:** Reduce average content production cycle-time by $\geq 70\%$.
2. **G2 — Bilingual quality:** Deliver publish-grade Arabic and English from a single workflow, with translation parity.
3. **G3 — Brand integrity:** Guarantee every published asset inherits approved brand tokens and guidelines.
4. **G4 — Governance:** Enforce who-can-do-what with a complete, queryable audit trail.
5. **G5 — Sovereignty:** Keep all content and AI processing under HUMAIN's control, in-Kingdom.

4. Business Impact and Value

4.1 Quantified impact (target outcomes)

Metric	Baseline (typical)	Target with Create Studio	Impact
Time to produce a bilingual page	2–4 days	2–4 hours	~85% faster
Translation turnaround (EN⇌AR)	1–2 days per asset	Minutes (AI draft + human review)	~90% faster
SEO metadata coverage	Manual / inconsistent	Auto-generated for 100% of content	Full coverage
Brand-compliance review effort	High, manual	Guided by system, exception-only	Major reduction
Tooling consolidation	3–5 separate tools	1 platform	Lower TCO + simpler ops
Content reuse	Low (copy-paste)	Block-based, versioned, reusable	Higher consistency

4.2 Strategic value

- **Sovereign capability** — a national-grade content platform owned and operated in-Kingdom, not rented from abroad.
- **Scalable team output** — small teams produce at large-team volume; specialists focus on judgement, not mechanics.
- **Compounding brand equity** — every asset is consistent by construction, strengthening brand over time.
- **Defensible governance** — auditable, role-gated workflows suitable for regulated and high-visibility communications.

5. Scope

5.1 In scope

- Prompt-first authoring (Create Studio) and structured CMS administration.
- Bilingual EN/AR content with RTL support.
- AI assistance: drafting, translation, summarisation, SEO, tagging, headline variants, FAQ generation, image alt-text, marketing copy.
- Block-based page composition and reusable content blocks.
- Editorial workflow, role-based access, versioning and audit logging.
- Brand and design-token management.

- Semantic + Arabic-aware search.
- Media management and AI-assisted/asynchronous video rendering.
- Secure delivery of published content to web channels.

5.2 Out of scope (this phase)

- Paid media / ad-buying automation.
- Native mobile applications (content is delivered to responsive web).
- Third-party social scheduling (available as a future integration).

6. Personas and Roles

The platform is **role-governed**. Each persona sees only what their role permits. The model is **7 roles realising 9 personas** — the same role is specialised into distinct personas by **ABAC scope attributes** (site, department, and content locale) rather than by minting a separate role per scope. This keeps the permission surface small and auditable while still giving each persona a precise remit.

Persona	Role	Scope (ABAC)	Primary jobs-to-be-done
Platform Administrator	admin	Global, unrestricted	Configure platform, manage users and roles, oversee all content and settings
Site Administrator	siteAdmin	Site-scoped (sites)	Own a specific site/brand surface end-to-end — create, review and publish within that site; cannot manage users or global settings
Author (EN)	author	Locale = EN	Draft and edit English content with AI assistance
Author (AR)	author	Locale = AR	Draft and edit Arabic content with AI assistance
Reviewer	reviewer	Site-scoped	Move content through editorial gates; quality-check
Publisher	publisher	Site-scoped	Approve and publish to live channels
Brand Manager	brand	Global brand assets	Own brand guidelines and design tokens
Department Author (e.g. HR)	author	Department-scoped (department)	Produce department content within guardrails (e.g. HR owns Careers)
Viewer	viewer	Read-only	Read-only access to published content and dashboards

The **7 roles** are admin , siteAdmin , author , reviewer , publisher , brand , and viewer . The **9 personas** above are these roles combined with scope attributes — e.g. *Author (EN)* and *Author (AR)* are both the author role distinguished by their **locale** scope, and *Department Author* is author

distinguished by its **department** scope. `siteAdmin` is the one genuinely distinct authority introduced for the Site Administrator persona: site-scoped admin powers (create / review / publish within its sites) without platform-level user, role, or global-settings control, so it cannot escalate privileges.

7. Functional Requirements

7.1 Create Studio — Prompt-First Authoring

- **FR-1** The system **shall** let a user generate publish-ready drafts from a natural-language brief.
- **FR-2** The system **shall** provide guided AI actions on any content: *summarise, translate, generate SEO, suggest headlines, generate FAQs, write alt-text, draft marketing copy, extract tags*.
- **FR-3** The system **shall** preview content live in both English and Arabic, including correct RTL layout.
- **FR-4** The system **shall** present a unified workspace covering Projects, Content, Brand, Design System, and Search.

7.2 Structured Content Management

- **FR-5** The system **shall** model content as governed collections (pages, articles, blog posts, FAQs, sites, media, brand guidelines, users).
- **FR-6** The system **shall** compose pages from **reusable, versioned content blocks**: Hero, Rich Text, Media, CTA, Cards, FAQ, Stats, Quote, Gallery, Embed, Features, Steps, and Logos.
- **FR-7** The system **shall** autosave drafts and retain a version history per document, enabling rollback.
- **FR-8** Content **shall** be scoped to a *site*, so teams only touch the surfaces they own (attribute-based access).

7.3 Bilingual and Localisation

- **FR-9** Every content field **shall** support per-locale values (EN/AR) with independent draft/published state.
- **FR-10** The system **shall** provide AI-assisted EN⇌AR translation that preserves meaning and tone, always followed by human review.
- **FR-11** The delivery layer **shall** render Arabic right-to-left with correct typography.
- **FR-11a** The **platform UI itself shall be switchable between English and Arabic** — one click, with full right-to-left layout and Arabic typography — independently of the content locale.

7.4 Editorial Workflow and Governance

- **FR-12** Content **shall** progress through an editorial gate: **Draft** → **In Review** → **Approved**, with only authorised roles able to advance it.
- **FR-13** Publishing **shall** be permission-gated; unauthorised users cannot push content live.
- **FR-14** Every create/update/delete/publish action **shall** be recorded in an immutable **audit log**.

7.5 Brand and Design System

- **FR-15** The system **shall** maintain centrally managed **brand guidelines** and **design tokens** (colours, type, spacing) shared by all surfaces.

- **FR-16** The system **shall** offer AI-assisted brand-guideline suggestions to accelerate brand setup and evolution.

7.6 Intelligent Search

- **FR-17** The system **shall** provide search across all content, including **Arabic-aware** and **semantic (meaning-based)** retrieval.

7.7 Media and Video

- **FR-18** The system **shall** manage media assets with AI-generated descriptive alt-text for accessibility.
- **FR-19** The system **shall** support **asynchronous video rendering** with status tracking.

7.8 Agent and Integration Surface

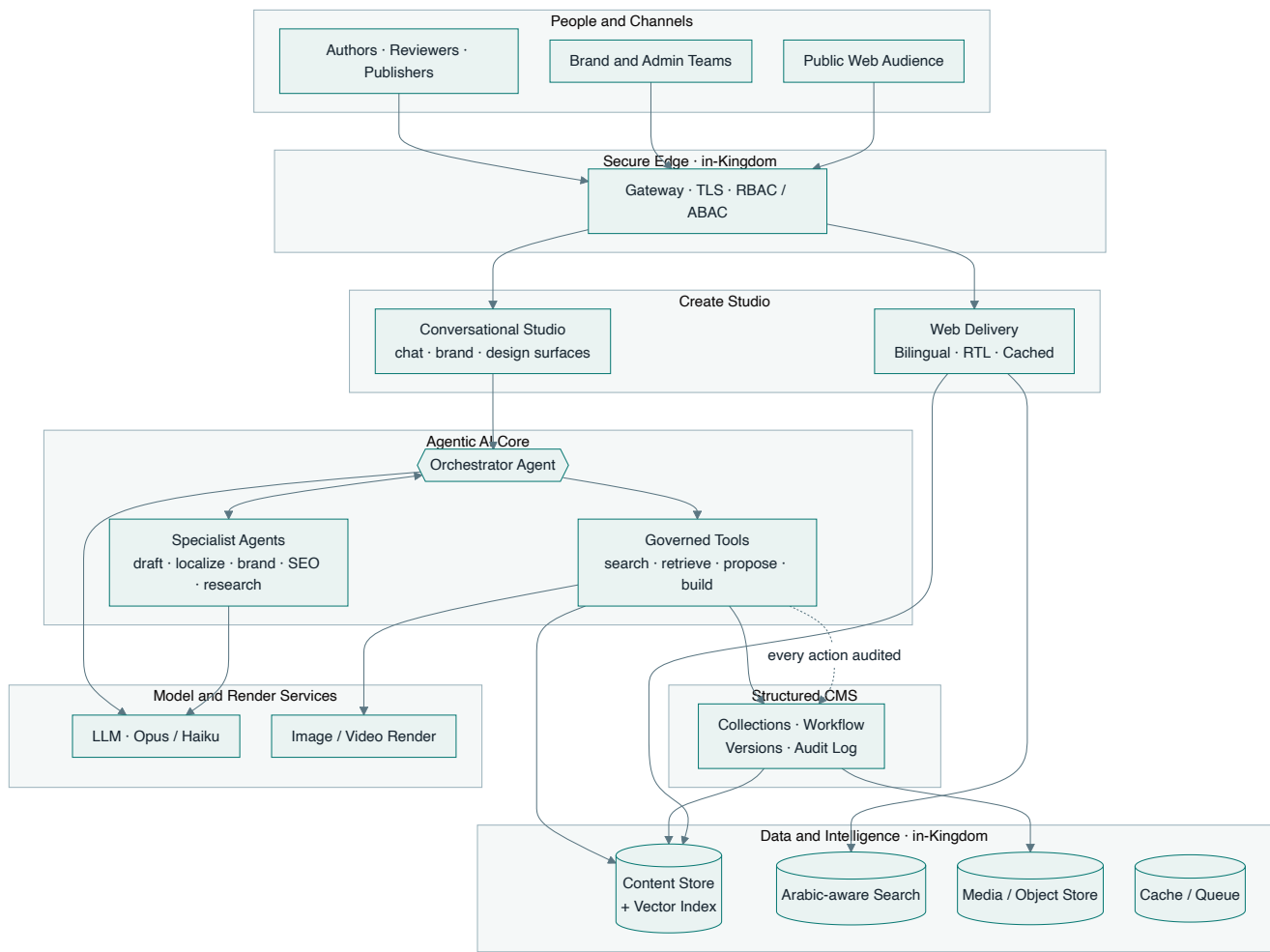
- **FR-20** The system **shall** expose a governed, audited tool interface (search / retrieve / propose content) for approved AI agents and downstream systems.

8. Non-Functional Requirements

Category	Requirement
Sovereignty	All content, data and AI processing hosted in-Kingdom (KSA) , under HUMAIN's control.
Security	Role-based + attribute-based access control; secure cookies; CORS/CSRF protection; secrets isolated from content.
Auditability	Complete, queryable audit trail of all content actions.
Localisation	First-class EN + AR (RTL); Arabic-aware search and translation.
Performance	Sub-second authoring interactions; cached, incrementally-rendered public delivery.
Scalability	Horizontally scalable services; queue-backed background processing for heavy jobs.
Reliability	Stateless application tier; managed data tier with health monitoring.
Extensibility	Provider-abstracted AI layer — model choice can evolve without re-platforming.
Observability	Centralised logs, metrics and tracing for operations.

9. Proposed Solution Architecture

9.1 High-level architecture



9.2 Logical components

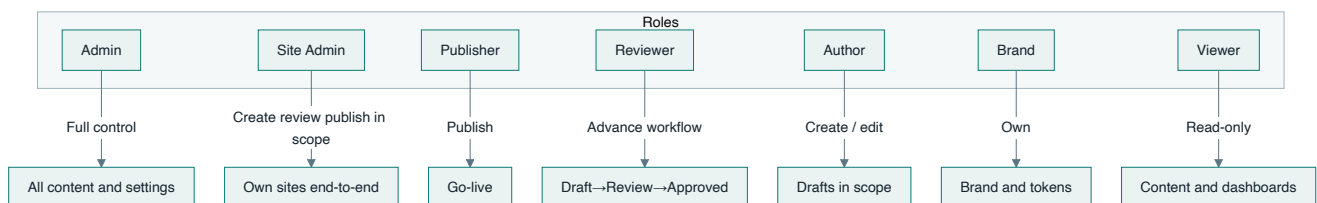
Layer	Responsibility
Create Studio	Natural-language authoring, AI actions, live bilingual preview
Structured CMS	Content schema, roles, editorial workflow, versioning, audit
AI Service	Drafting, translation, SEO, tagging, summarisation, semantic indexing
Web Delivery	Fast, cached, RTL-correct public rendering
Content Store	Single source of truth + vector embeddings for semantic search
Search Index	Arabic-aware and semantic retrieval
Cache / Queue	Performance + reliable background processing (e.g. video, indexing)
Object Store	Media and rendered assets, in-region

10. Content Lifecycle



Every transition is **role-gated** and **logged**.

11. Role and Permission Model



- **7 roles** → **9 personas**: `admin`, `siteAdmin`, `author`, `reviewer`, `publisher`, `brand`, `viewer` cover all nine personas (see §6); locale and department personas are the `author` role differentiated by ABAC scope.
- **Site scoping (ABAC)**: site admins, authors and editors act only on the sites assigned to them (empty = all).
- **Department scoping**: departmental authors are confined to their department's content.
- **No privilege escalation**: `siteAdmin` has site-scoped editorial authority but cannot manage users, roles or global settings — only `admin` can.

12. Agentic AI Platform

The platform's intelligence is delivered by a team of **specialised AI agents**, each owning a step of the content lifecycle and coordinated by an **orchestrator**. Agents are goal-directed and multi-step: given a brief or objective, they plan the work, retrieve relevant context, act through governed tools, check their own output, and hand off to one another until a publish-ready deliverable exists — with a human approving anything that goes live.

12.1 What makes it agentic

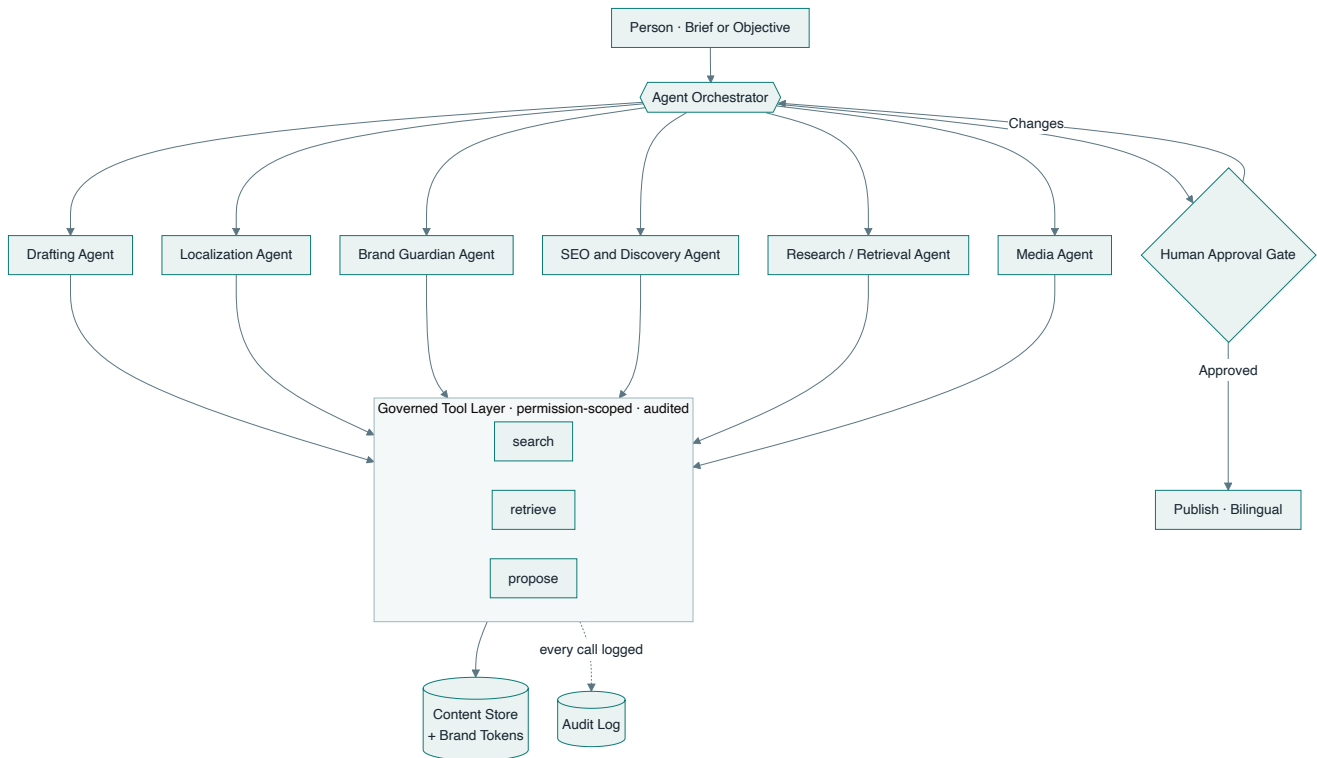
- **Goal-directed** — agents work from an objective ("produce a bilingual launch page from this brief"), not a single canned prompt.
- **Multi-step and self-checking** — they plan → retrieve → act → verify, iterating until the result meets brand and quality criteria.

- **Tool-using** — agents act only through a **governed tool layer** (search, retrieve, propose) rather than touching data directly.
- **Context-aware (RAG)** — they ground output in HUMAIN's own content and brand tokens via semantic, Arabic-aware retrieval.
- **Collaborative** — specialised agents hand off to each other under the orchestrator's plan.
- **Governed** — every agent runs under the same role/permission model as people, every action is audited, and **agents propose** — **humans dispose**.

12.2 Agent catalogue

Agent	What it does	Acts through
Orchestrator	Interprets the brief, plans the steps, delegates to specialist agents, assembles the final deliverable	Plan + governed tools
Drafting Agent	Turns a brief into a structured, block-based draft	<code>content_propose</code>
Localization Agent	Produces EN⇌AR translations with tone and meaning parity	<code>content_get</code> / <code>propose</code>
Brand Guardian Agent	Checks tone and visual tokens against brand guidelines; flags drift before review	Brand tokens + <code>content_get</code>
SEO and Discovery Agent	Generates metadata, keywords, taxonomy tags and internal-link suggestions	<code>content_search</code> / <code>propose</code>
Research / Retrieval Agent	Grounds output in HUMAIN's own content via semantic, Arabic-aware retrieval (RAG)	<code>content_search</code>
Media Agent	Generates descriptive alt-text and orchestrates image/video rendering	Media + render tools
Editorial Agent	Summarises changes, prepares review notes, routes content through the workflow gates	Workflow + <code>content_get</code>

12.3 Orchestration and governance



- Agents operate under the **same RBAC/ABAC** as human users — an agent can only touch content its scope permits.
- Agents **propose**; they **cannot publish**. A human approval gate stands before go-live.
- **Every agent tool call is written to the audit log**, giving a complete, queryable record of what each agent did and why.
- The agent runtime is **provider-abstracted** — HUMAIN is never locked to a single model vendor, and the strongest model can be assigned per task.

12.4 What the agent builds

From one conversation, the agent decides what is needed and **builds a complete, interactive, usable outcome** — not just a suggestion of copy:

Outcome	What the user gets
Website / landing page / app	A complete, responsive site built and previewed live in chat , with Open / Download
Structured content	Article, blog, page, press release, event, product, case study, FAQ, email, campaign or social post — an editable card that can be published to a CMS module (as a draft for review) or downloaded
Video	A production-ready script / storyboard with a one-click render to a real video
Image	A generated image , rendered inline
Brand guideline	A full guideline to verify, publish as the active brand the agents follow , or download
Design theme	A complete colour / type / radius theme with live preview, applied to the platform

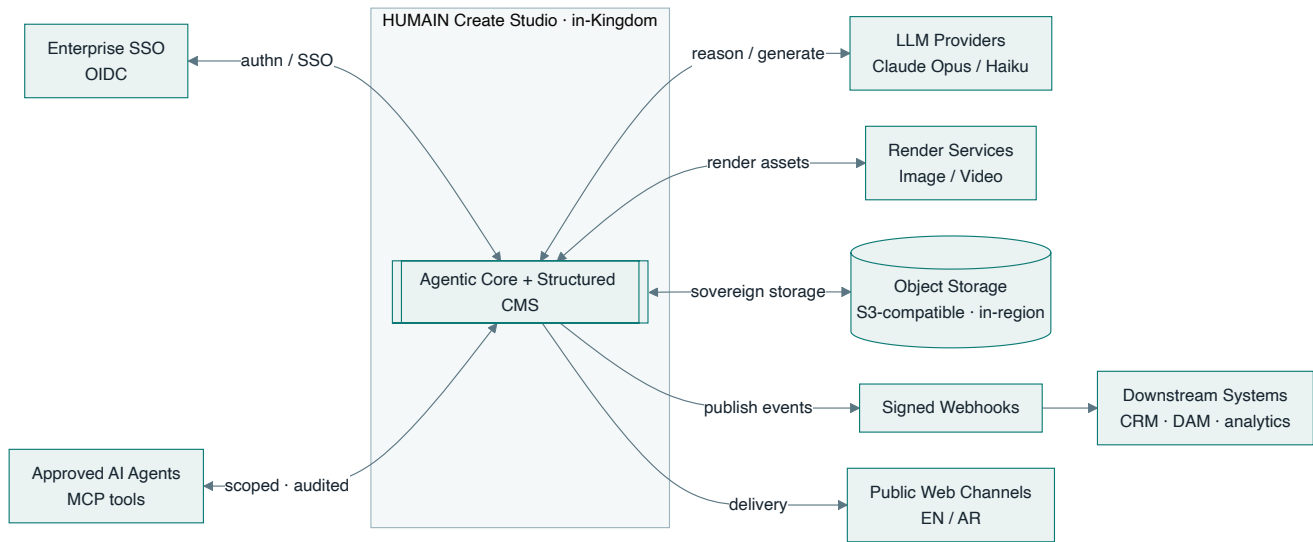
The agent draws on direct assist skills too — each also a one-click action: **drafting, EN⇌AR translation, SEO metadata, summarisation, headline variants, FAQ generation, taxonomy tagging, image alt-text and semantic, Arabic-aware search.**

Agents act; humans decide. No content reaches the public without human approval, and every agent action is permission-scoped and logged.

13. Integration Requirements

Integration	Purpose
Enterprise SSO (OIDC)	Single sign-on with HUMAIN identity
Object storage (S3-compatible, in-region)	Sovereign media and asset storage
Webhooks (signed)	Notify downstream systems on publish / content events
Agent tool interface	Governed programmatic access for approved AI agents
Web channels	Deliver published, bilingual content to public sites
LLM and render services	Provider-abstracted models (Opus/Haiku) + image/video rendering

13.1 Integration architecture

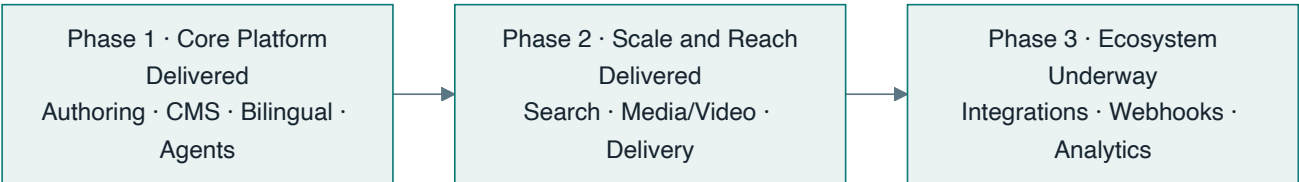


14. Security, Compliance and Sovereignty

- **In-Kingdom hosting** — data residency under HUMAIN's control.
- **Role-based + attribute-based access control** at every layer.
- **Editorial gating** prevents unreviewed or unauthorised publishing.
- **Immutable audit log** of all content actions for accountability.
- **Secrets isolation** — credentials and keys separated from content and code.
- **Secure transport** — encrypted (HTTPS/TLS) across all surfaces; secure session cookies; CORS/CSRF protections.

15. Delivery Roadmap and Status

The core programme (Phases 1–2) is **delivered and live** in-Kingdom; Phase 3 is **underway** (the agentic build set and MCP tool surface are already shipped).



Phase	Outcome	Focus	Status
Phase 1 — Core Platform	Teams author, translate and govern bilingual content	Studio, CMS, RBAC, workflow, agentic AI, brand tokens	Delivered
Phase 2 — Scale and Reach	Content is discoverable and richly delivered	Arabic-aware semantic search, media/video, cached public delivery	Delivered
Phase 3 — Ecosystem	Platform extends across HUMAIN's stack	Agent/MCP integrations, signed webhooks, advanced analytics	Underway

16. Success Metrics (KPIs)

KPI	Target
Bilingual content cycle-time reduction	≥ 70%
SEO metadata coverage	100% of published content
Brand-compliance exceptions at publish	Trending to near-zero
Translation review effort	≥ 80% reduction vs. manual
Audit completeness	100% of content actions logged
Author satisfaction (time-to-publish)	Measurable improvement quarter-on-quarter

17. Assumptions and Dependencies

- HUMAIN provides in-region infrastructure (or approves the proposed hosting) and an OIDC identity provider.
- Content owners define the initial brand guidelines and design tokens.
- AI model access is provisioned within HUMAIN's approved policy.
- Human review remains in the loop for all AI-assisted output prior to publish.

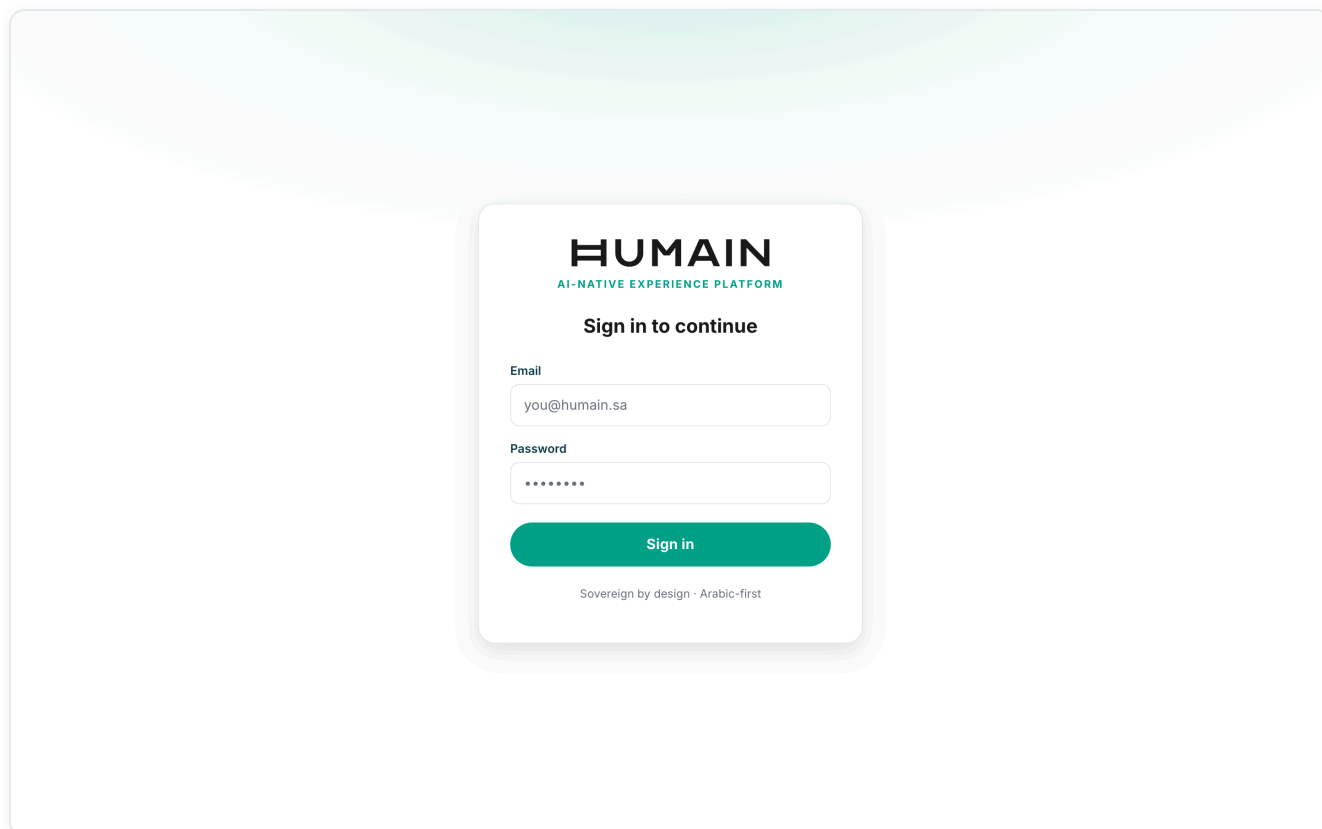
18. Glossary

Term	Meaning
Create Studio	The prompt-first authoring experience
Block	A reusable, versioned content component used to compose pages
Design tokens	Centralised brand values (colour, type, spacing) shared across surfaces
RTL	Right-to-left layout, required for Arabic
RBAC / ABAC	Role-based / attribute-based access control
Semantic search	Meaning-based retrieval beyond keyword matching
Audit log	Immutable record of all content actions
In-Kingdom	Hosted and processed within Saudi Arabia for data sovereignty

Appendix A — Product Snapshots

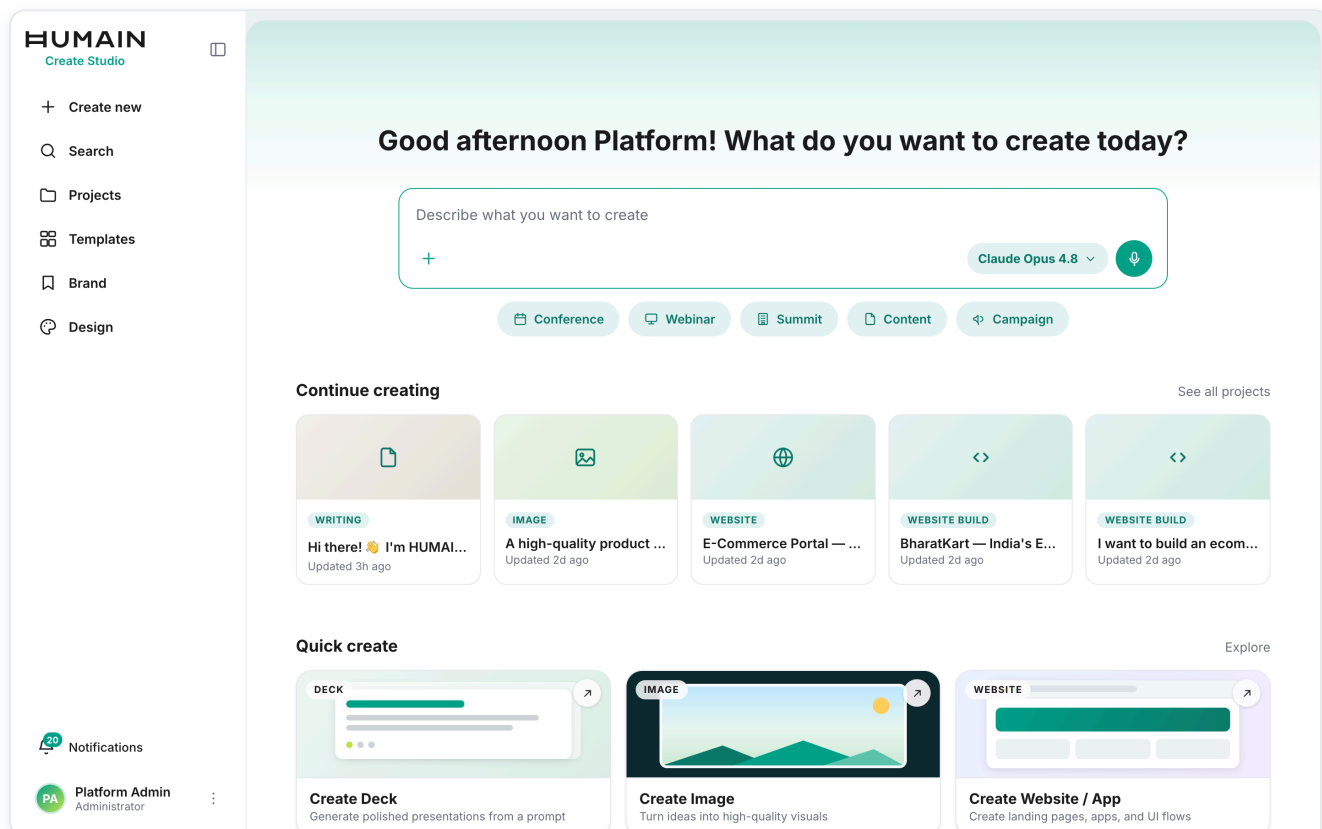
The following snapshots are taken from the working HUMAIN Create Studio platform, showing the realized capabilities described above.

A.1 Secure bilingual sign-in



Branded, HTTPS sign-in backed by role-based access (OIDC-ready). Every session is permission-scoped.

A.2 Create Studio — one prompt box for everything



A single conversational entry point. The agent decides what to build; recent work and quick-create paths sit alongside.

A.3 Conversational, on-brand assistant

The screenshot displays the HUMAIN Create Studio interface. On the left is a sidebar with navigation options: 'Create new', 'Search', 'Projects', 'Templates', 'Brand', and 'Design'. Below these are 'Notifications' and a user profile for 'Platform Admin Administrator'. The main area features a chat window with a light green header. A prompt asks, 'What can you help me create? Give me a short rundown.' The assistant's response is: 'Hey! Great to meet you. I'm your creative studio — I can take an idea from a quick brief all the way to a polished, ready-to-use outcome. Here's the short rundown:'. Below this is a table titled 'What I can build for you'.

What you want	What you get
Websites & landing pages	A live, working page you can preview right here
Content	Articles, blog posts, pages, press releases, case studies, FAQs, product copy, emails, campaigns, social posts — as editable, publishable cards
Video	A production-ready script/storyboard with one-click render (ads, teasers, reels)
Images	Illustrations, posters, graphics, visuals — rendered inline
Brand identity	Full brand guidelines you can edit, publish or download
Themes	Colour palettes, fonts and design systems with live swatches

Below the table, the assistant lists 'A few things worth knowing':

- **Bilingual** — I work in **English and Arabic** (or both side by side).
- **Grounded & on-brand** — I can pull from existing content and apply brand guidelines so everything stays consistent.
- **Interactive** — whatever I make, you can refine it, and I'll give you next-step options.

Next, it asks 'Want to start? Just tell me what you're working on. For example:' and provides three example prompts:

- "Build a landing page for a new AI product launch"
- "Write a LinkedIn post announcing our partnership"
- "Design a brand identity for a fintech startup"

It then asks 'What's on your mind?' and shows 'Copy', 'Retry', and 'Claude Opus 4.8' options. At the bottom is a text input field with a plus icon and a dropdown menu set to 'Claude Opus 4.8'.

Natural, streaming replies (not form dumps) with rich formatting — the assistant explains what it can build, recommends next steps, and works bilingually (EN/AR), grounded in the active brand.

A.4 Build a website — live, in-chat preview

The screenshot displays the HUMAIN Create Studio interface. On the left is a sidebar with navigation options: 'Create new', 'Search', 'Projects', 'Templates', 'Brand', and 'Design'. Below these are 'Notifications' and a user profile for 'Platform Admin Administrator'. The main workspace shows a conversation with an AI agent. The agent's response includes a 'WEBSITE' section with a description of a landing page for the 'HUMAIN AI Summit' in Riyadh, featuring a cinematic design, agenda, speaker grid, and registration passes. It also lists three tasks: 'Translate to Arabic', 'Add real speakers & sponsors', and 'Wire up registration'. Below the text is a 'LIVE PREVIEW' of the website. The preview shows a dark-themed landing page with the HUMAIN logo, the event title 'Where Enterprise Meets Intelligence', and a call to action 'Reserve Your Seat'. At the bottom of the chat area is a text input field with a plus icon and a dropdown menu set to 'Claude Opus 4.8'.

HUMAIN
Create Studio

+ Create new
Q Search
📁 Projects
📄 Templates
🔖 Brand
🕒 Design

🔔 Notifications
PA Platform Admin Administrator

Build a high-impact landing page now for the HUMAIN AI Summit in Riyadh, audience enterprise executives, in English. Do not ask questions — build it.

WEBSITE

Your **HUMAIN AI Summit** landing page is live in the preview — a cinematic, executive-grade site with the dark-ink and teal-to-lime brand treatment, a conversion-focused hero, full 2-day agenda, speaker grid, and three-tier registration passes.

A few high-impact things I can do next:

- **Translate to Arabic** — ship a full bilingual (EN/AR) version with RTL layout for the Gulf audience.
- **Add real speakers & sponsors** — drop in actual names, titles, headshots, and partner logos to replace placeholders.
- **Wire up registration** — connect the passes to a working form/checkout, or add an email-capture + countdown timer to drive urgency.

Want me to start with any of these — or tweak the hero copy, colours, or section order?

LIVE PREVIEW Open Download

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RIYADH · 2025

Where Enterprise Meets Intelligence

The HUMAIN AI Summit convenes the world's most influential enterprise and government leaders to turn AI ambition into measurable, production-grade outcomes — at sovereign scale.

Reserve Your Seat → View Agenda

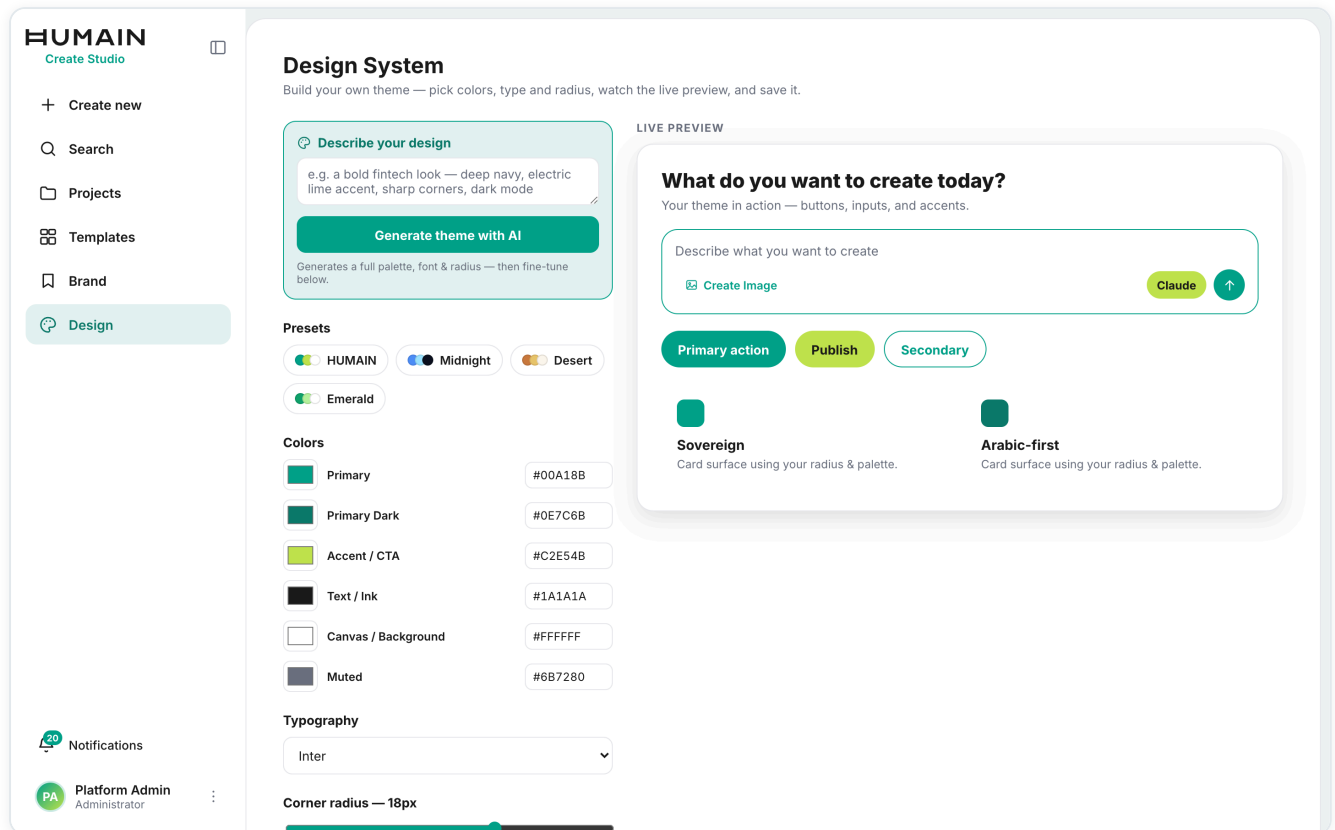
Copy ⌂ Retry Claude Opus 4.8

Ask a follow-up or refine...

+ Claude Opus 4.8

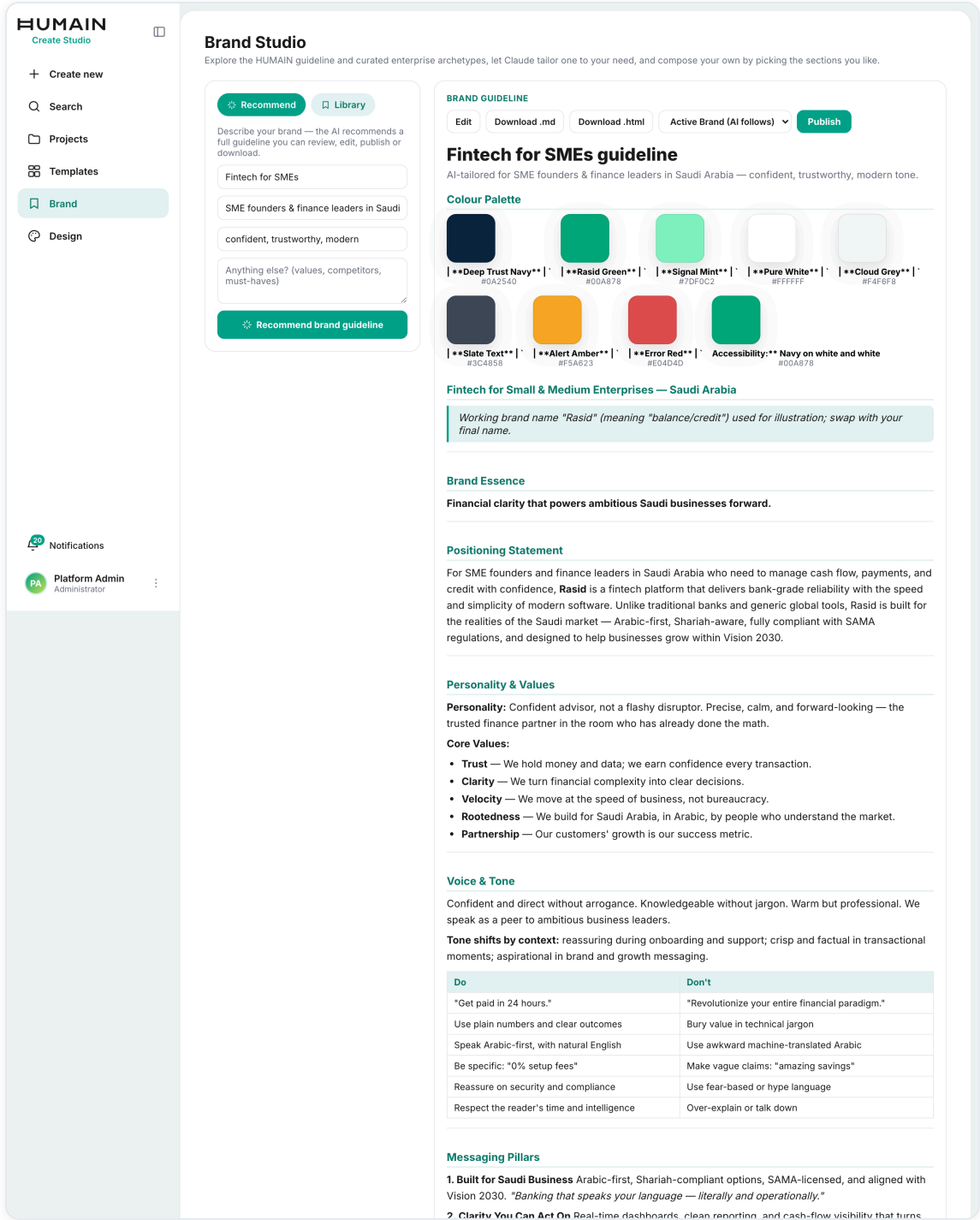
Ask for a site and the agent **builds it**, rendered live in the conversation with Open / Download — not a description. The same pattern produces editable, publishable content, images and video.

A.5 Design System — prompt-driven, agent-governed theme



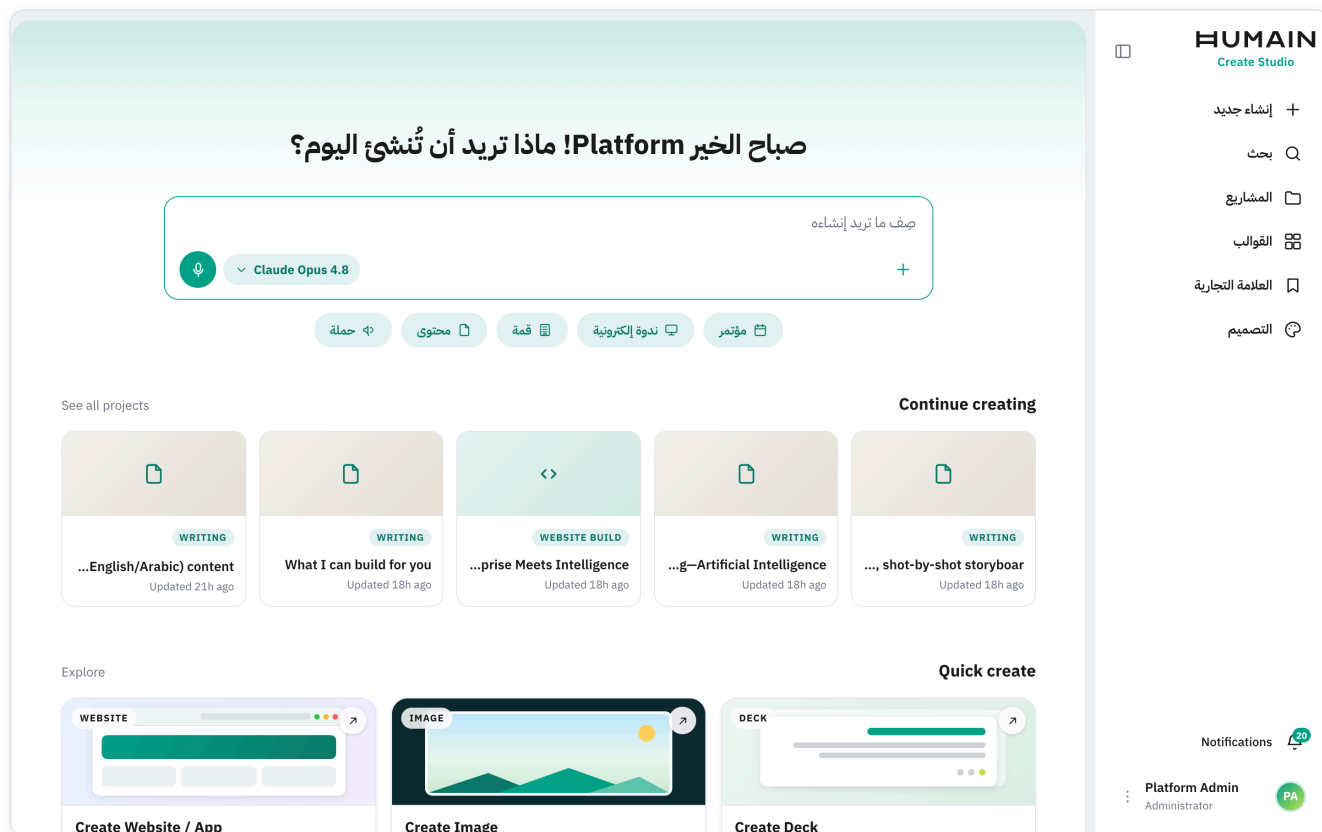
Describe a look and the agent generates a complete theme (palette, type, radius, light/dark) with a live preview; tokens are shared across every surface, EN + AR.

A.6 Brand Studio — recommend → review → publish



The AI recommends a full brand guideline to review and verify, then publish as the **active brand the agents follow**, or download.

A.7 Bilingual platform UI — English / Arabic



A one-click language switch turns the entire platform UI to **Arabic with full right-to-left layout** (and back to English) — navigation, prompts and actions all localised. The same content store serves both languages.

Part II — UI/UX and User Journeys

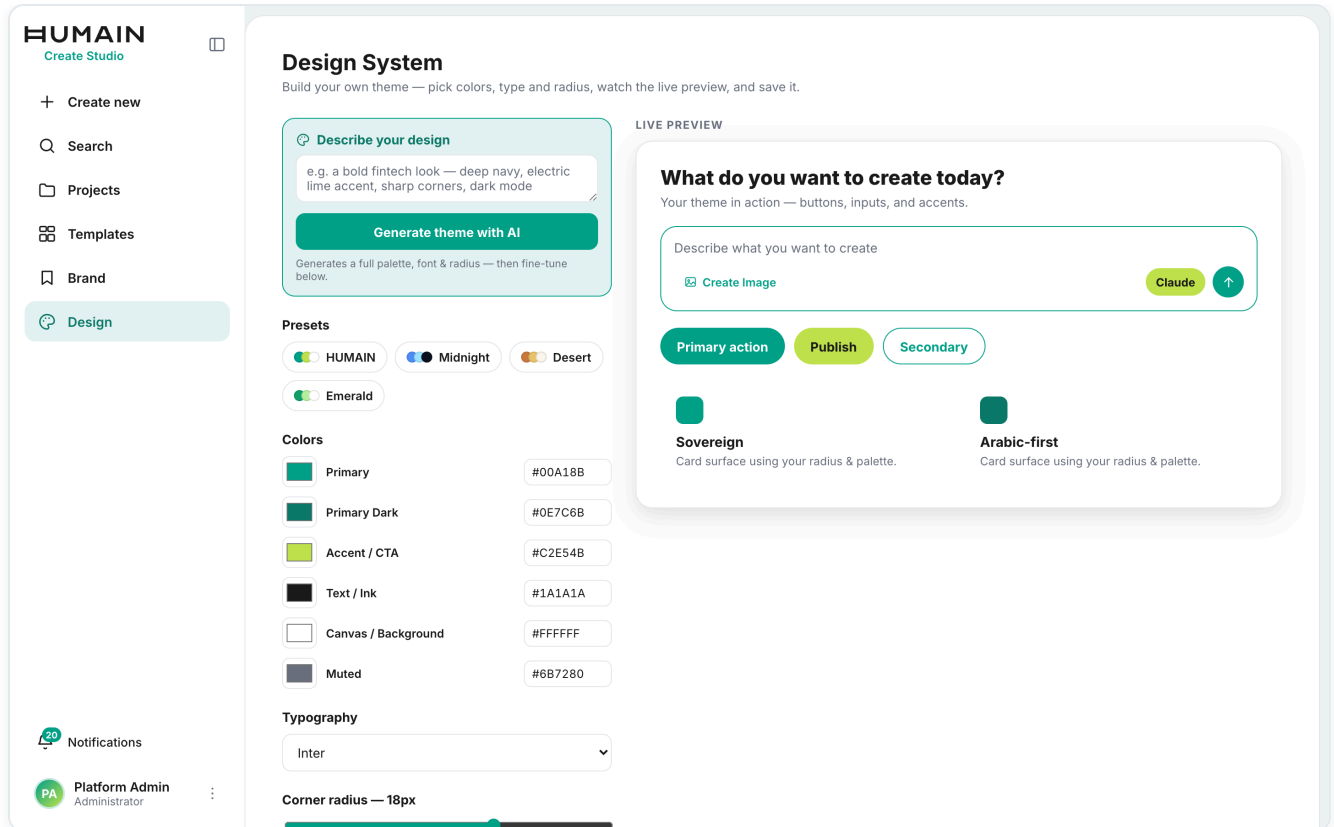
1. Design Principles

The experience is built on six principles, each visible in the product:

- **Agent-first, not form-first.** A person describes intent in natural language; the agent decides what to build and produces an interactive result. Forms are the fallback, never the starting point.
- **Conversational, like Claude.** Replies stream in as natural messages with light actions (Copy, Retry) — not boxed reports. The assistant asks a clarifying question when needed and recommends next steps.
- **Always an interactive outcome.** Every capability returns something the user can *use* — a live website preview, an editable/publishable card, a theme to apply — never a dead-end text dump.
- **Bilingual by design.** English and Arabic are first-class, including right-to-left layout; the type system carries both (Inter + IBM Plex Sans Arabic).
- **On-brand by construction.** Every surface inherits the central brand tokens; the agent grounds output in the active brand.
- **Governance you can see.** Roles, review gates, publish-as-draft and a complete audit trail are part of the flow — agents propose, humans decide.

2. Visual Design System

A single token system (colour, typography, spacing, radius) drives every surface, in light and dark, EN and AR. It is itself **prompt-driven** — a user can describe a look and the agent generates a complete, accessible theme with a live preview.



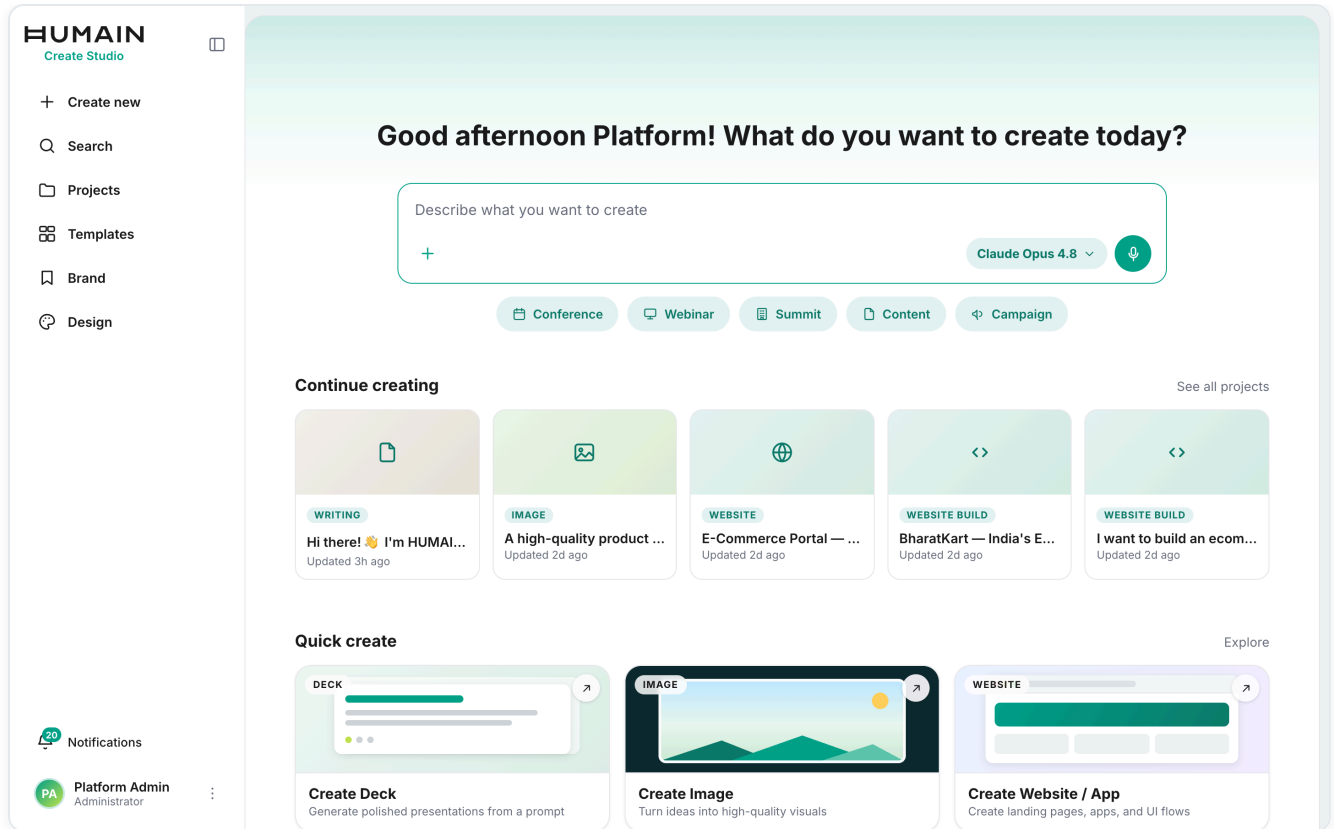
- **Palette:** primary teal, primary-dark, lime accent, ink, canvas, muted — each with usage.
- **Typography:** Inter (Latin) + IBM Plex Sans Arabic (Arabic), shared scale.
- **Shape and surface:** configurable corner radius; light/dark surfaces; generous whitespace.

3. Navigation and Information Architecture

A persistent left sidebar gives one-click access to every capability; the workspace fills the rest.

- **Create new** — start a fresh conversation.
- **Search** — find across content (semantic, Arabic-aware).
- **Projects** — everything produced, as reusable assets.
- **Templates** — structured content modules.
- **Brand** — Brand Studio (recommend → publish brand guidelines).
- **Design** — Design System (prompt-driven theme).

The Studio home is a **single prompt box**: the user types intent and the agent routes to the right outcome — no need to pick a tool first.



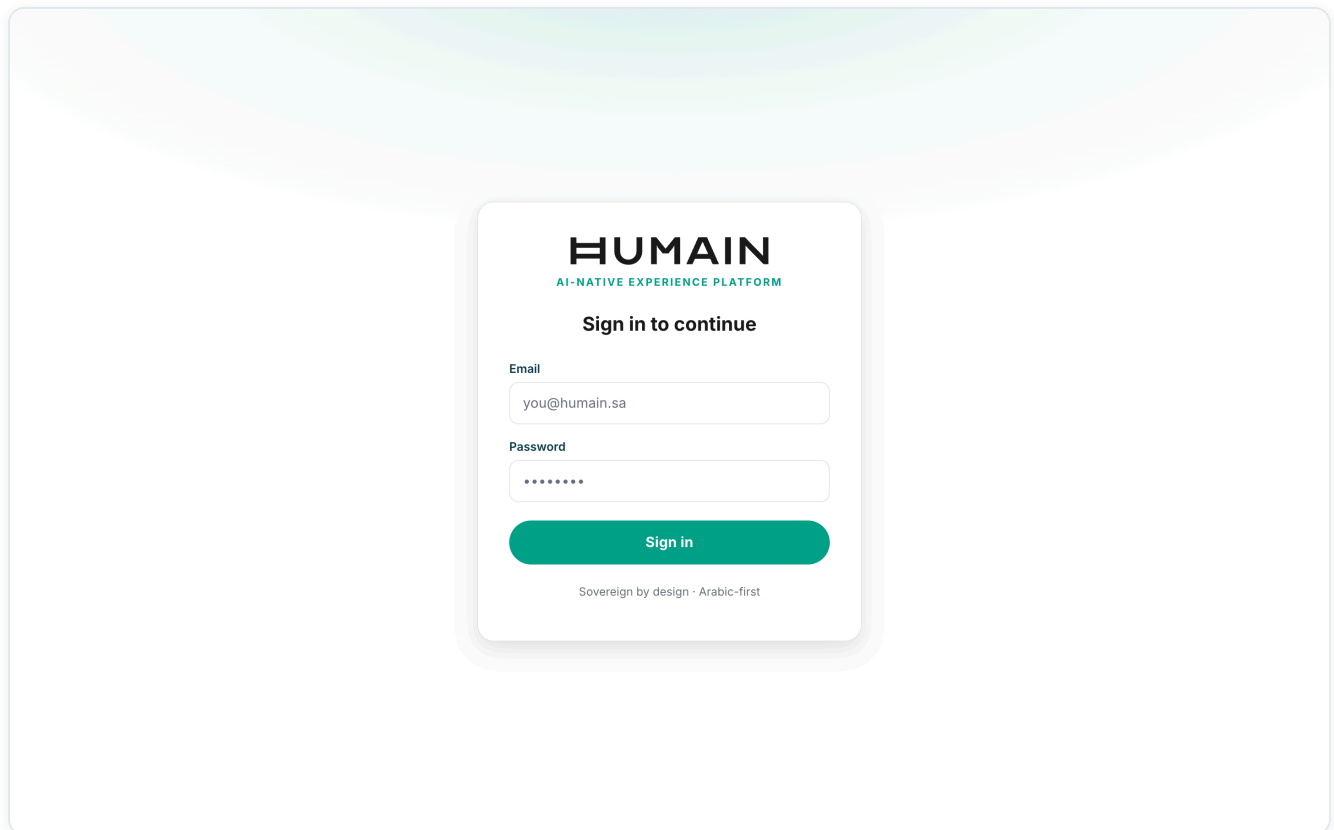
4. Core Interaction Patterns

- **Conversation thread** — user messages (right) and natural assistant replies (left), streamed live.
- **Artifact cards** — when the agent builds something, it appears inline as an interactive card: a website preview, a content card, a brand card, a theme.
- **Action row** — under each reply: Copy, Retry, and the model used.
- **Manual override** — every generated artifact can be edited by hand (the human is never locked out).
- **Publish / Download** — outcomes can be published (to a CMS module or as the active brand, as a draft for review) or downloaded.

5. User Journeys

Each journey below is shown as it appears in the product, with the step-by-step UI/UX.

5.1 Sign in

A screenshot of the HUMAIN sign-in interface. The interface is centered on a light blue background with a subtle wave pattern. It features a white card with the HUMAIN logo at the top, followed by the tagline 'AI-NATIVE EXPERIENCE PLATFORM'. Below this is the heading 'Sign in to continue'. There are two input fields: 'Email' with the placeholder 'you@humain.sa' and 'Password' with masked characters. A green 'Sign in' button is positioned below the password field. At the bottom of the card, the text 'Sovereign by design · Arabic-first' is displayed.

HUMAIN
AI-NATIVE EXPERIENCE PLATFORM

Sign in to continue

Email

you@humain.sa

Password

Sign in

Sovereign by design · Arabic-first

Journey: 1) The user opens the branded sign-in over HTTPS. 2) Enters email + password (OIDC-ready for SSO). 3) On success a secure, role-scoped session starts and they land in Create Studio — seeing only what their role permits.

5.2 Converse with the assistant

The screenshot displays the HUMAIN Create Studio interface. On the left is a sidebar with navigation options: 'Create new', 'Search', 'Projects', 'Templates', 'Brand', and 'Design'. Below these are 'Notifications' and a user profile for 'Platform Admin Administrator'. The main area features a chat window with a light blue header containing the prompt 'What can you help me create? Give me a short rundown.' The assistant's response is a friendly greeting followed by a table titled 'What I can build for you'. The table lists various services like websites, content, video, images, brand identity, and themes. Below the table, the assistant provides 'A few things worth knowing' and 'Want to start?' suggestions. At the bottom, there's a text input field with a plus icon and a dropdown menu set to 'Claude Opus 4.8' with a microphone icon.

HUMAIN
Create Studio

+ Create new
Search
Projects
Templates
Brand
Design

Notifications
Platform Admin Administrator

What can you help me create? Give me a short rundown.

Hey! Great to meet you. I'm your creative studio — I can take an idea from a quick brief all the way to a polished, ready-to-use outcome. Here's the short rundown:

What I can build for you

What you want	What you get
Websites & landing pages	A live, working page you can preview right here
Content	Articles, blog posts, pages, press releases, case studies, FAQs, product copy, emails, campaigns, social posts — as editable, publishable cards
Video	A production-ready script/storyboard with one-click render (ads, teasers, reels)
Images	Illustrations, posters, graphics, visuals — rendered inline
Brand identity	Full brand guidelines you can edit, publish or download
Themes	Colour palettes, fonts and design systems with live swatches

A few things worth knowing

- **Bilingual** — I work in **English and Arabic** (or both side by side).
- **Grounded & on-brand** — I can pull from existing content and apply brand guidelines so everything stays consistent.
- **Interactive** — whatever I make, you can refine it, and I'll give you next-step options.

Want to start? Just tell me what you're working on. For example:

- "Build a landing page for a new AI product launch"
- "Write a LinkedIn post announcing our partnership"
- "Design a brand identity for a fintech startup"

What's on your mind?

Copy ⌂ Retry Claude Opus 4.8

Ask a follow-up or refine...

+ Claude Opus 4.8 🎤

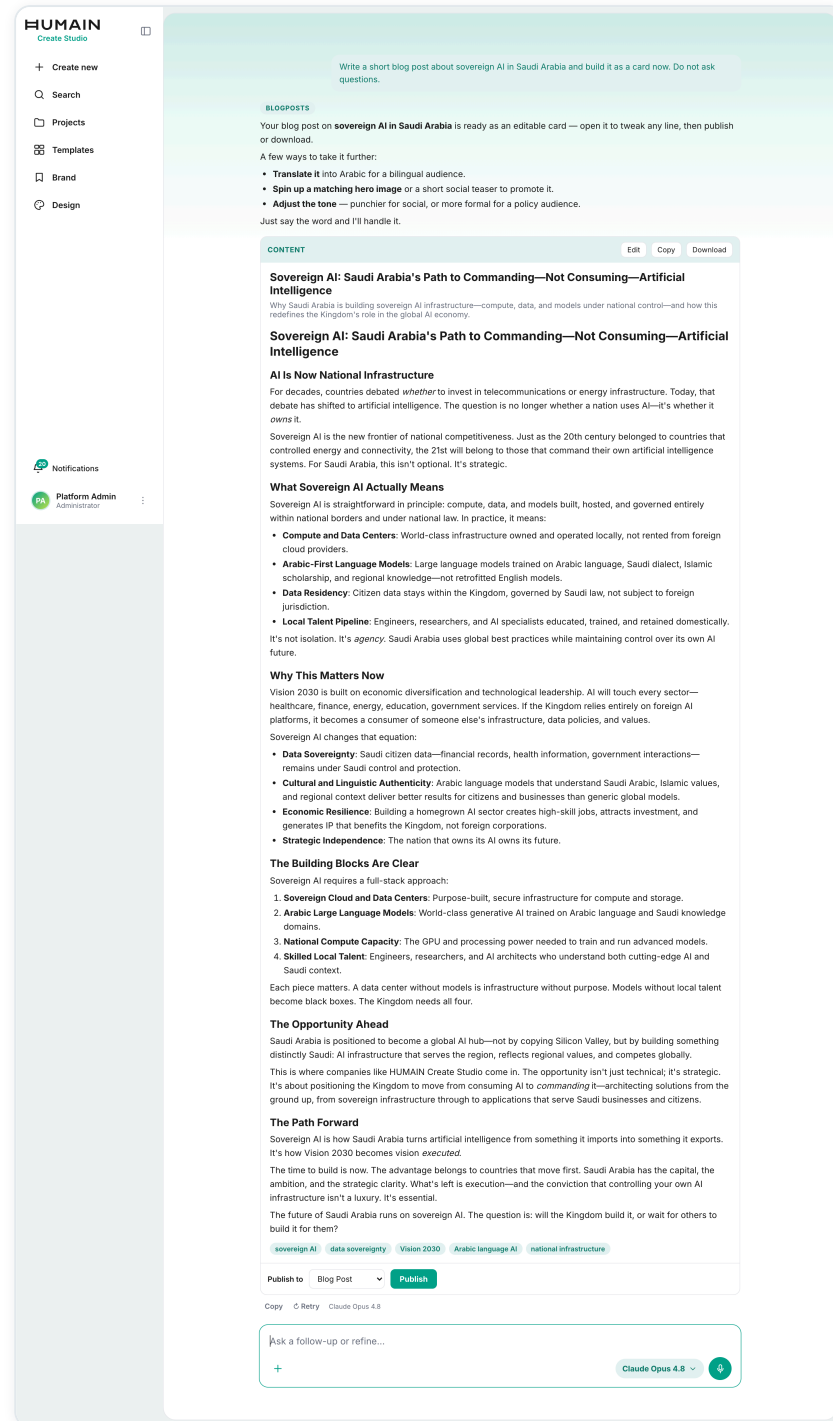
Journey: 1) The user types a question or a greeting. 2) The assistant **replies conversationally** (it does not over-build) — here it explains what it can create, in a richly formatted, scannable message. 3) It recommends concrete next steps and invites a follow-up. 4) The user can Copy or Retry the reply. *UX note: simple input → conversation, not a form dump.*

5.3 Build a website

The screenshot displays the HUMAIN Create Studio interface. On the left is a sidebar with navigation options: 'Create new', 'Search', 'Projects', 'Templates', 'Brand', and 'Design'. Below these are 'Notifications' and a user profile for 'Platform Admin Administrator'. The main workspace shows a 'WEBSITE' section with a text prompt: 'Build a high-impact landing page now for the HUMAIN AI Summit in Riyadh, audience enterprise executives, in English. Do not ask questions — build it.' Below this, a paragraph states: 'Your HUMAIN AI Summit landing page is live in the preview — a cinematic, executive-grade site with the dark-ink and teal-to-lime brand treatment, a conversion-focused hero, full 2-day agenda, speaker grid, and three-tier registration passes. A few high-impact things I can do next:' followed by a bulleted list: 'Translate to Arabic', 'Add real speakers & sponsors', and 'Wire up registration'. Below the list is a question: 'Want me to start with any of these — or tweak the hero copy, colours, or section order?'. A 'LIVE PREVIEW' section shows a mobile view of the website with a dark header, the HUMAIN logo, and a hero section titled 'Where Enterprise Meets Intelligence' with a subtitle 'The HUMAIN AI Summit convenes the world's most influential enterprise and government leaders to turn AI ambition into measurable, production-grade outcomes — at sovereign scale.' and two buttons: 'Reserve Your Seat' and 'View Agenda'. At the bottom of the preview are 'Copy', 'Retry', and 'Claude Opus 4.8' options. Below the preview is a text input field 'Ask a follow-up or refine...' with a plus icon and a dropdown menu set to 'Claude Opus 4.8'.

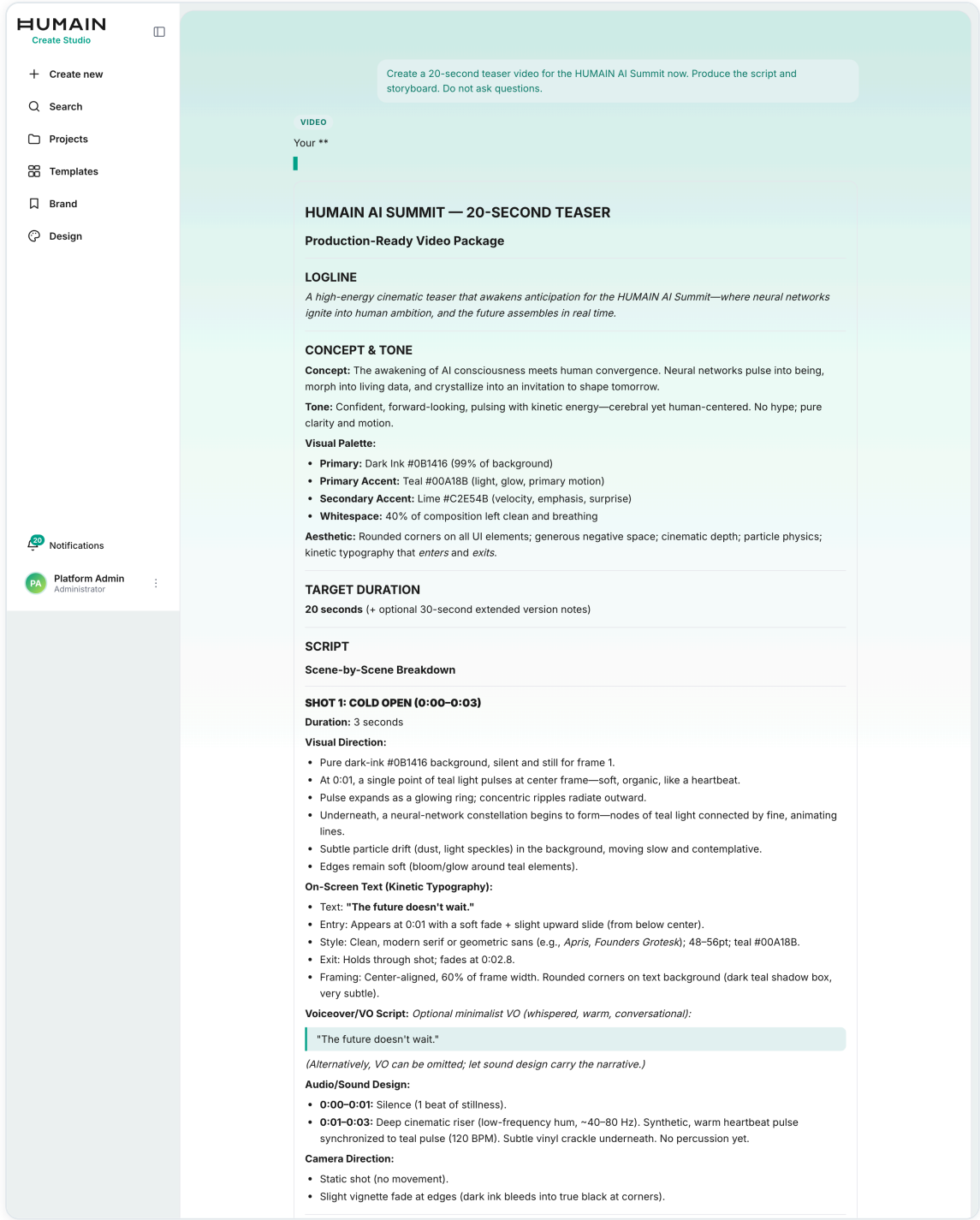
Journey: 1) The user asks for a site ("build a launch page for our summit"). 2) If details are missing the agent asks one quick question; otherwise it **builds it**. 3) The finished site renders **live in the conversation** with **Open** and **Download**. 4) The agent introduces it and suggests edits ("translate to Arabic", "add a pricing section"). 5) A follow-up refines the same site in place.

5.4 Create and publish content



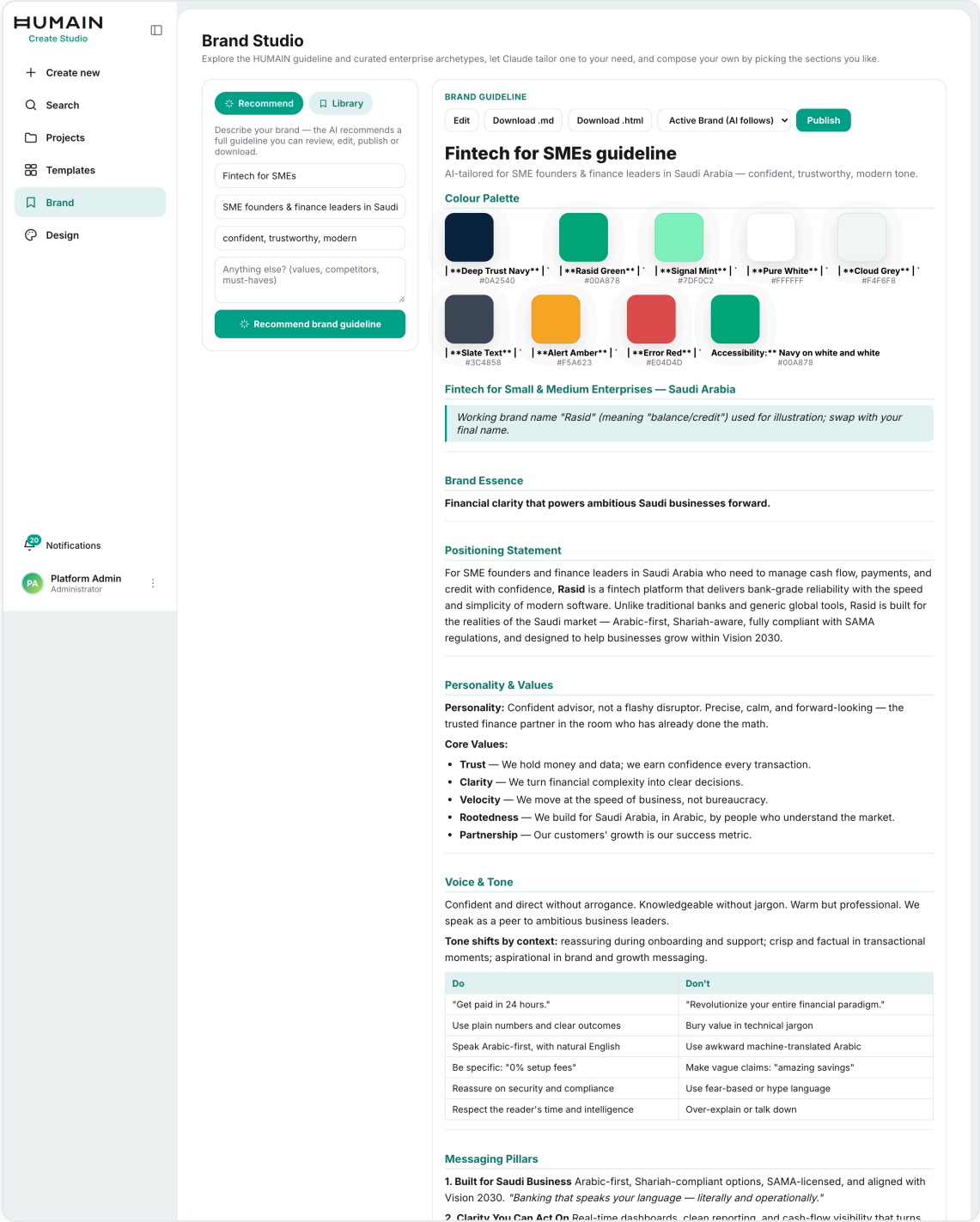
Journey: 1) The user asks for content (blog post, article, press release, FAQ, email...). 2) The agent returns an **editable content card** — title, body (rich), summary, auto-tags. 3) The user can **Edit** (manual override), **Copy** or **Download**. 4) They choose a destination module from the **Publish to** dropdown and publish — it lands as a **draft for review** (agents propose, humans approve).

5.5 Generate a video



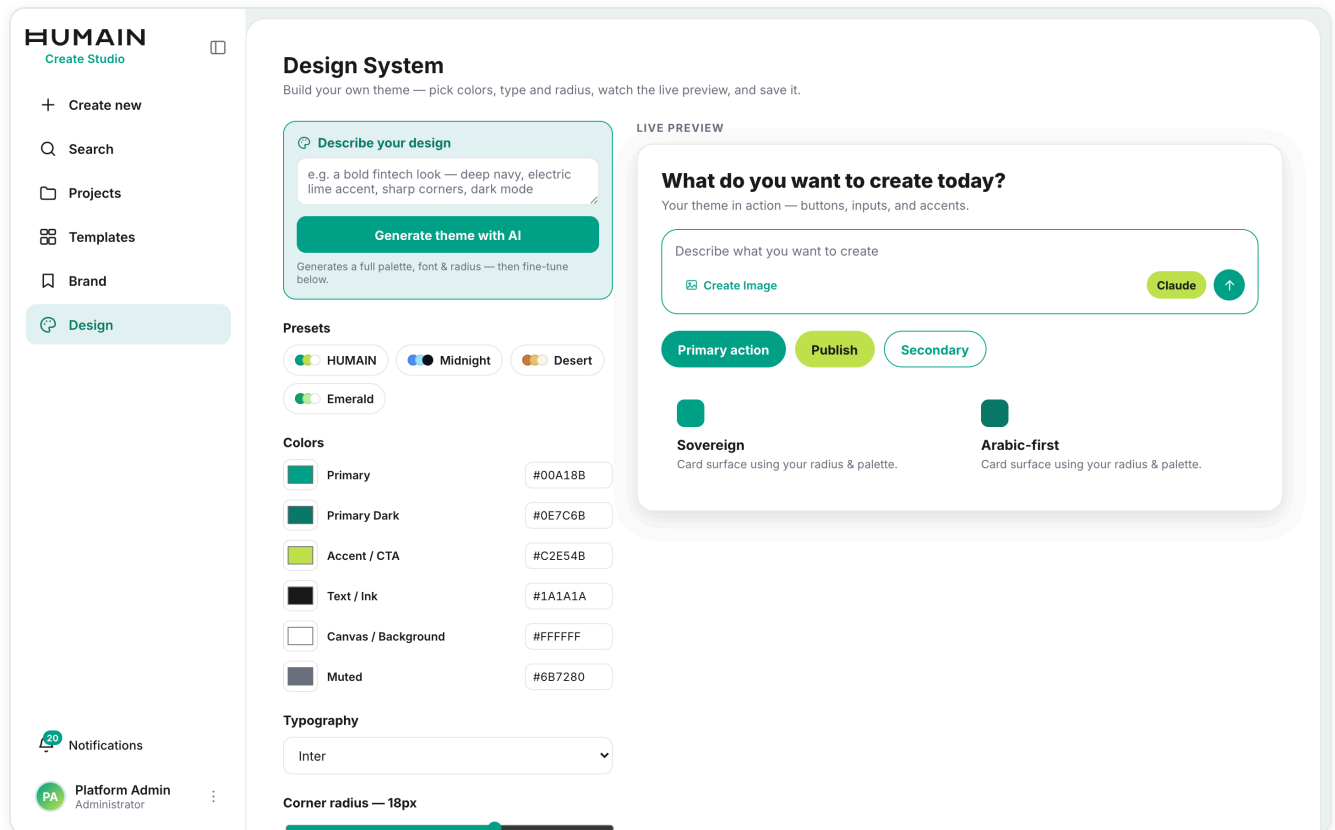
Journey: 1) The user asks for a video/teaser. 2) The agent produces a **production-ready package** — logline, concept, scene-by-scene script, shot list, storyboard. 3) A one-click **Render** control turns the script into a real video. 4) The user can refine tone, length or scenes in a follow-up.

5.6 Recommend and activate a brand



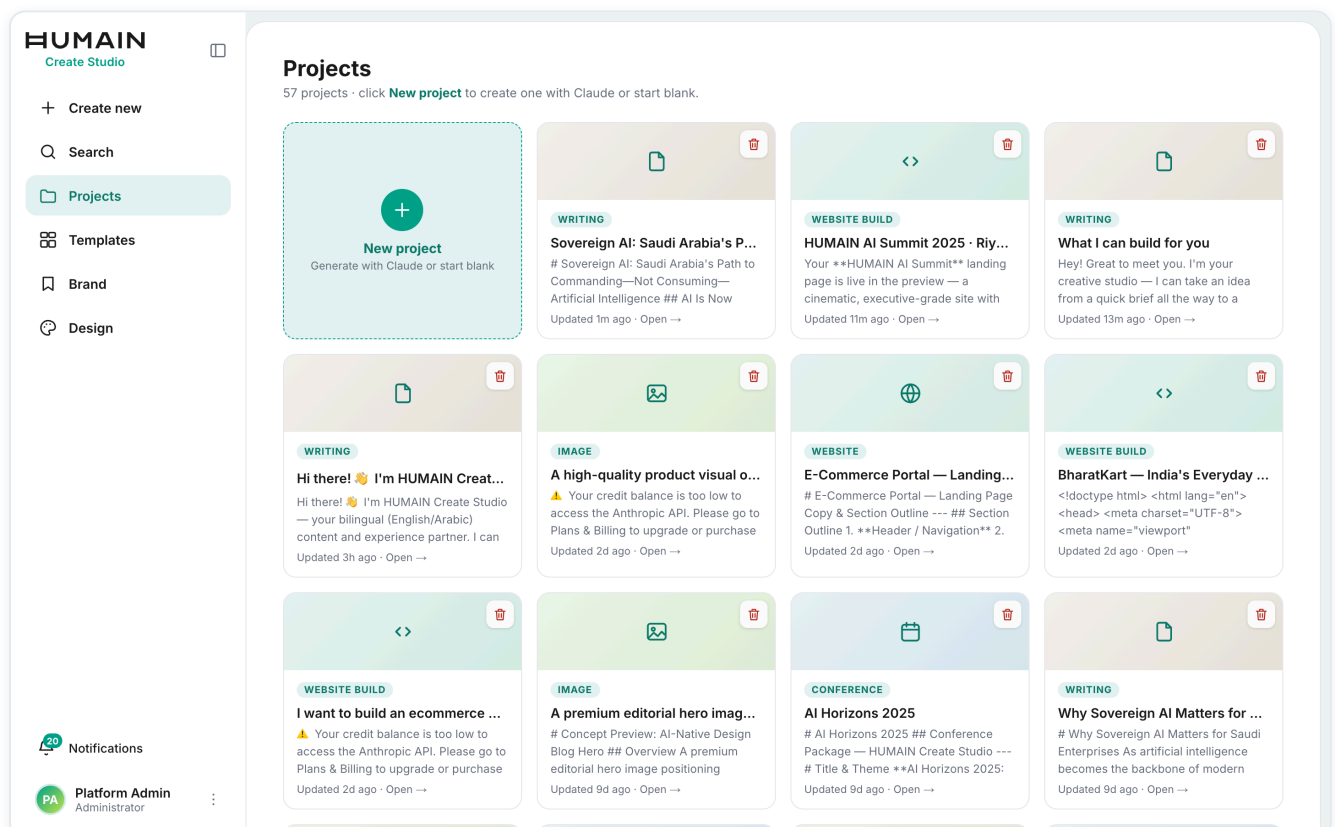
Journey: 1) The user describes the brand (industry, audience, tone) — or opens an archetype from the Library. 2) The AI **recommends a full guideline** (essence, voice, palette, typography, messaging). 3) The user **reviews and verifies**, editing any section. 4) They **Publish** it — as the **active brand the agents follow**, or to the library — or **Download** it.

5.7 Generate a design theme



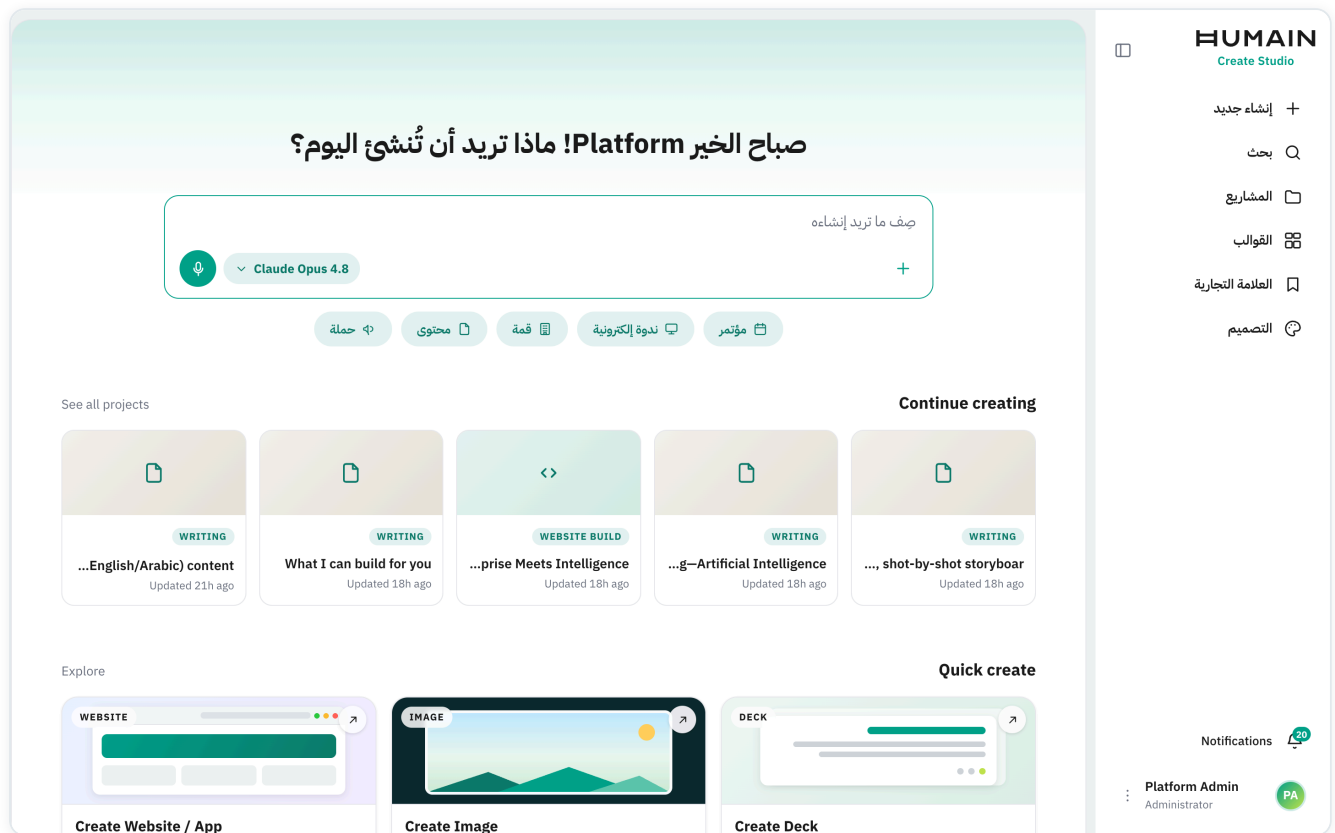
Journey: 1) In **Describe your design**, the user types a vibe ("bold fintech — deep navy, electric lime, dark mode"). 2) The agent generates a **complete theme** (palette, font, radius, light/dark). 3) The **live preview** updates instantly; the user fine-tunes any token by hand. 4) **Apply** saves it to site settings.

5.8 Manage projects



Journey: 1) Everything the user creates is saved as a **Project** asset. 2) They browse, search and reopen prior work. 3) Each project is typed (website, content, brand, video...) and ready to promote into the CMS.

5.9 Switch the platform language (English / Arabic)



Journey: 1) From the sidebar user menu the user picks **English** or 2) **العربية**. The entire platform UI switches instantly — selecting Arabic flips everything to **right-to-left** with the Arabic type system. 3) Navigation, greeting, prompt, quick actions and menus are localised. 4) The choice persists across sessions.

6. Accessibility and Localization

- **EN + AR with correct RTL** across authoring and delivery.
- **Bilingual typography** baked into the token system.
- **Keyboard + voice input** on the prompt box; readable contrast in light and dark.
- **Descriptive alt-text** generated for media by default.

7. Roles and Responsibilities

Persona	What they see and can do
Author (English, Arabic)	Create and edit drafts in scope, with full agent assistance
Reviewer	Advance content through the editorial gate
Publisher	Approve and publish to live channels
Brand Manager	Own the brand guidelines and the active brand
Admin	Configure the platform, users and settings, with full visibility
Viewer	Read-only access to content and dashboards

Every action a persona takes is permission-scoped and written to the audit log.

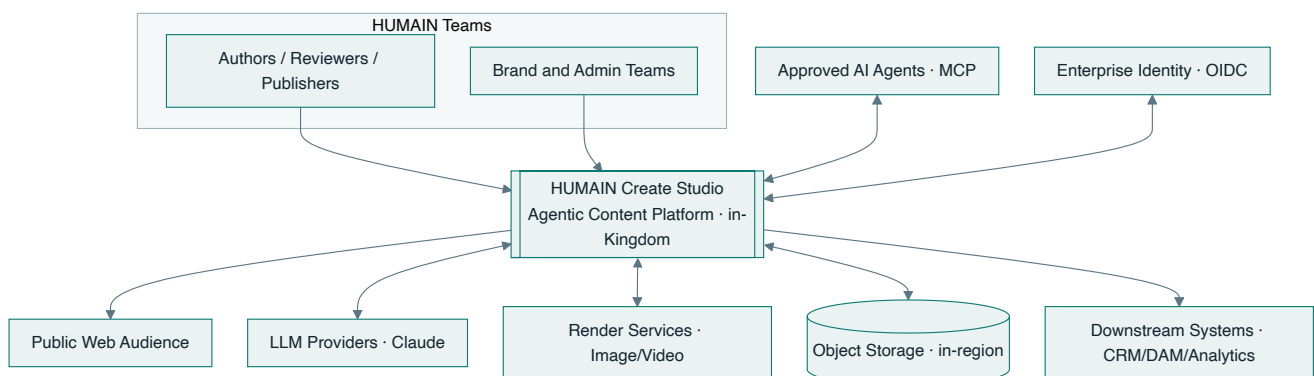
Part III — High-Level Architecture

1. Overview

HUMAIN Create Studio is an **agentic, AI-native content platform**. A team of specialised AI agents, coordinated by an orchestrator, turns natural-language intent into interactive, on-brand, bilingual outcomes — websites, content, brand guidelines, themes, images and video — all governed by role-based permissions, editorial gates and a complete audit trail, hosted in-Kingdom.

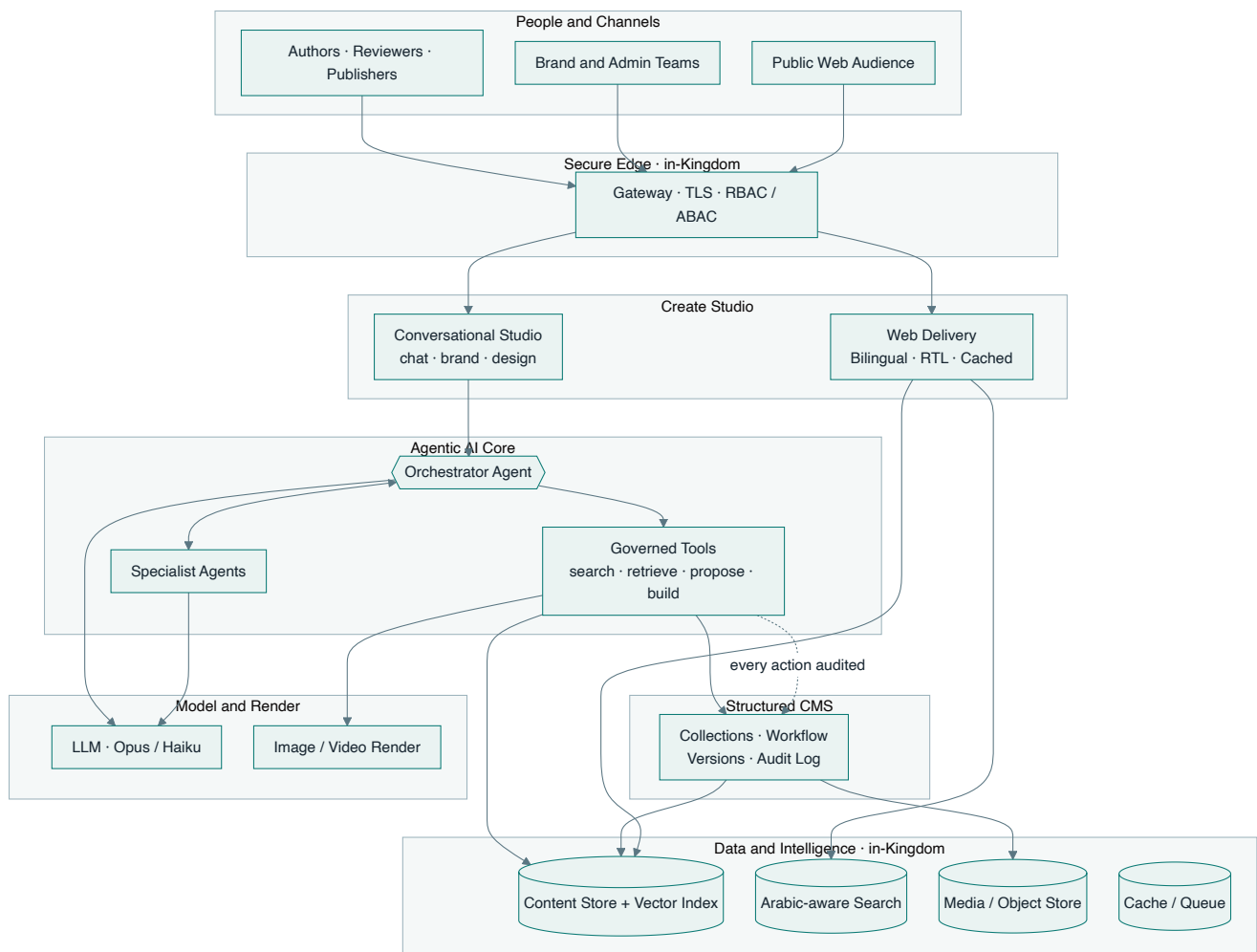
This document describes the system at a high level: context, components, the agentic core, data and integration architecture, request flow, deployment, and the non-functional posture.

2. System Context



The platform is the system-of-record for content; identity, models, render and storage are integrated services.

3. High-Level Architecture

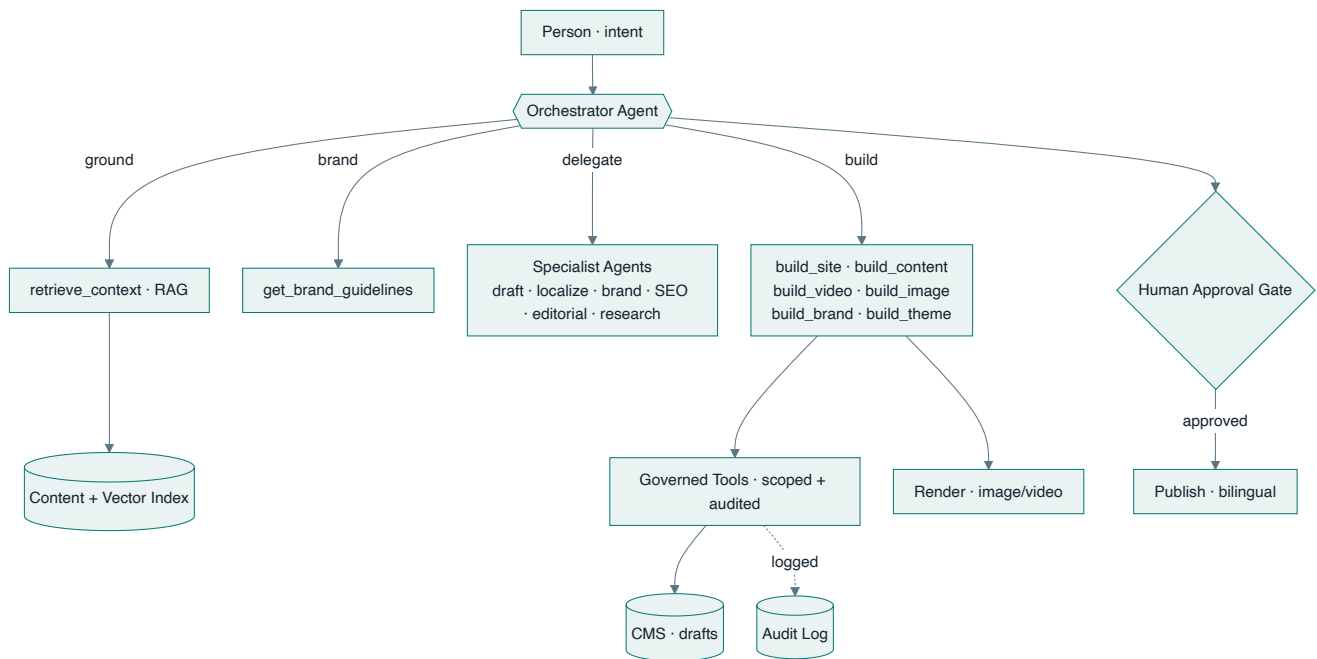


4. Component View

Component	Responsibility	Notes
Edge / Gateway	TLS termination, routing, auth, RBAC/ABAC	In-Kingdom; HTTPS only
Create Studio (console)	Conversational UI + Brand and Design surfaces	Streams agent output (SSE)
Agentic AI Core (AI service)	Orchestrator + specialists + governed tools	Tool-use loop on Claude
Structured CMS	Schema, RBAC/ABAC, editorial workflow, versions, audit	System-of-record
Web Delivery	Fast, cached, RTL-correct public rendering	Bilingual EN/AR
Content Store	Source of truth + vector embeddings	Postgres + pgvector
Search	Arabic-aware + semantic retrieval	—
Object Store	Media and rendered assets	S3-compatible, in-region
Cache / Queue	Performance + background jobs	Redis
Model and Render	LLM completion + image/video rendering	Provider-abstracted

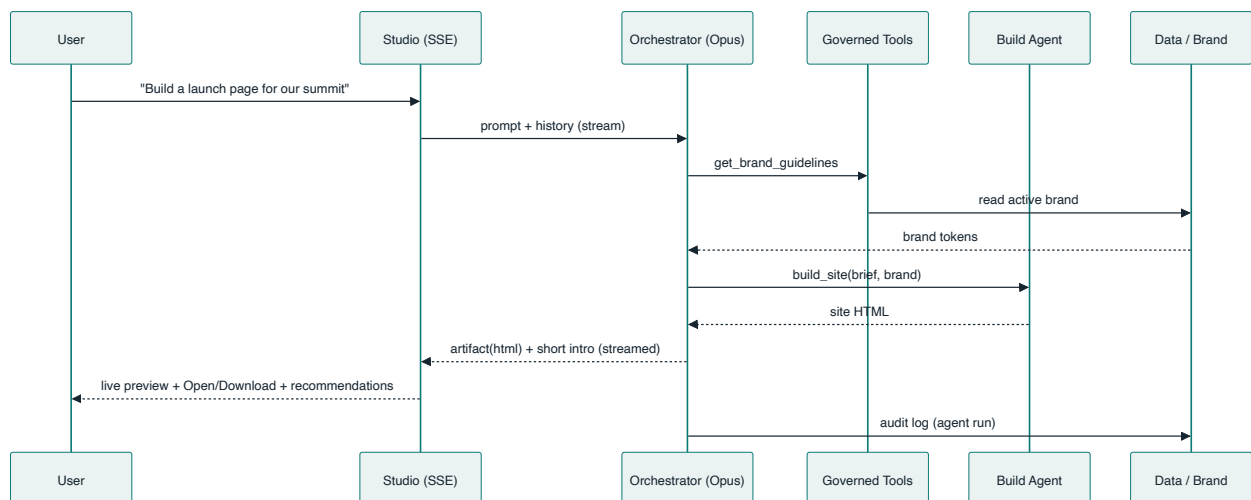
5. Agentic Core

The intelligence is an **orchestrator agent** running a multi-step tool-use loop. It plans, grounds itself in HUMAIN content and brand, delegates to specialist agents, and builds interactive outcomes — never a raw dump. Every tool call is audited; agents propose drafts, humans publish.



- **Orchestrator** runs on the strongest model (Opus) for reasoning/routing.
- **Specialists and generation** run on a fast model (Haiku) for cost efficiency.
- **Governed tools** are the only path to data — read (search/retrieve) and propose (drafts); never direct, unscoped writes.

6. Request Flow — "Build a website"



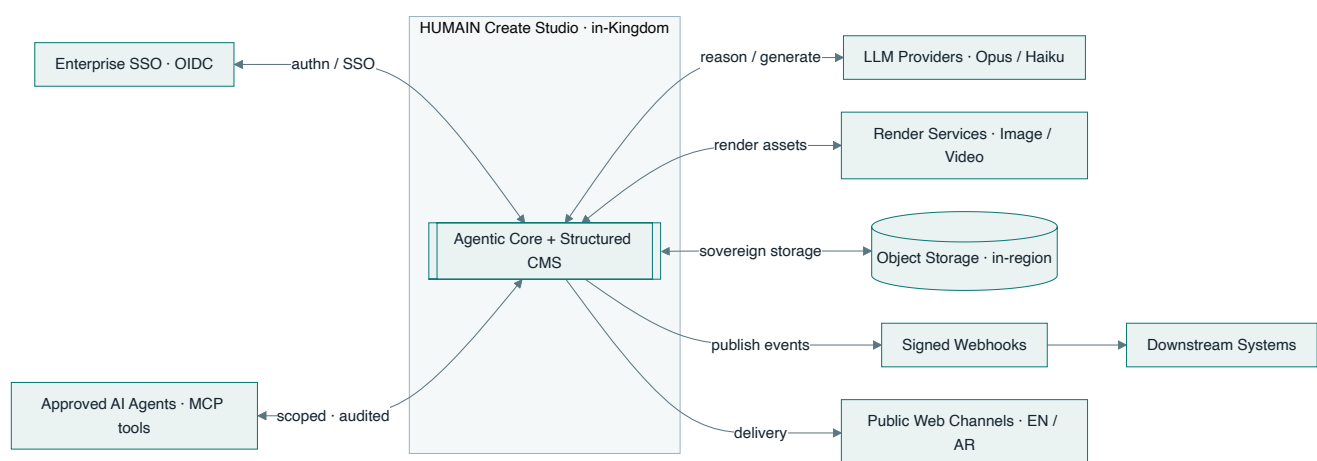
The same pattern serves every capability — only the build step differs (content, video, image, brand, theme).

7. Data Architecture

- **Content Store (Postgres):** the source of truth — collections, drafts, versions, workflow state, audit log.
- **Vector Index (pgvector):** sovereign, local embeddings (multilingual, 384-dim) power semantic, Arabic-aware retrieval (RAG) — no external embedding calls.
- **Search:** Arabic-aware + semantic retrieval over indexed content.
- **Object Store (S3-compatible, in-region):** media and rendered assets.
- **Cache / Queue (Redis):** performance and reliable background processing (indexing, render polling).

All content and embeddings remain **in-Kingdom**.

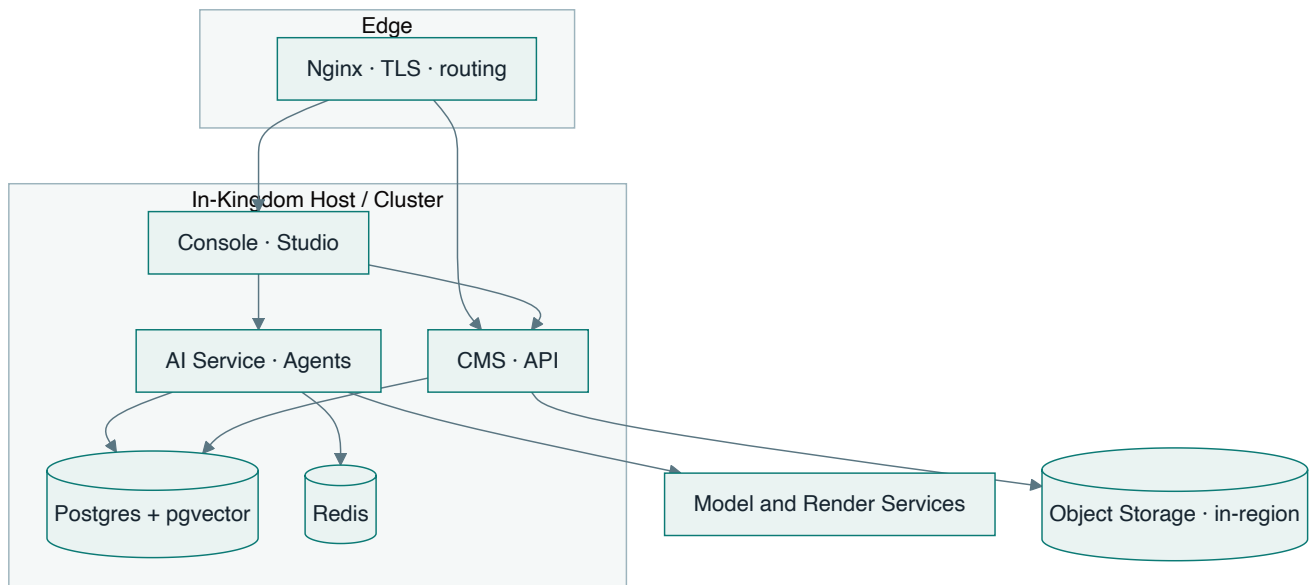
8. Integration Architecture



- **Provider-abstracted models** — HUMAIN is never locked to a single vendor.
- **MCP tool surface** — approved external agents act through the same governed, audited tools.
- **Signed webhooks** notify downstream systems on publish/content events.

9. Deployment Architecture

A self-contained, containerised stack, in-Kingdom, behind a TLS edge.



- **Stateless app tier** (console, CMS, AI service) — horizontally scalable.
- **Managed data tier** (Postgres+pgvector, Redis) with health checks.
- **One image, many services** — each container runs a different workload from a shared build.
- **Repeatable deploys** with image cache-busting per release and forced container recreation.

10. Security, Governance and Sovereignty

- **In-Kingdom** hosting and processing — data residency under HUMAIN's control.
- **RBAC + ABAC** at every layer (role, site-scope, department, locale).
- **Editorial gating** — Draft → In Review → Approved; agents propose drafts, humans publish.
- **Immutable audit log** of all content *and* agent actions.
- **Secrets isolation** — credentials/keys separated from content and code.
- **Secure transport** — HTTPS/TLS everywhere; secure session cookies; CORS/CSRF protection.
- **Human-in-the-loop** — nothing reaches the public without approval.

11. Technology Stack

Layer	Technology
Console / Web	Next.js (App Router), React, SSE streaming, EN/AR RTL
CMS	Payload (headless), RBAC/ABAC, versioned drafts, audit
AI service	Node/Fastify, Anthropic Claude (Opus + Haiku), tool-use loop
Agents / tools	Orchestrator + specialists; MCP governed tool surface
Embeddings	Local multilingual model (sovereign, 384-dim)
Data	Postgres + pgvector; Arabic-aware search; Redis
Render	Image/Video via provider-abstracted services
Storage	S3-compatible object storage, in-region
Edge / Infra	Nginx + TLS; containerised; one image, many services

12. Scalability, Reliability and Observability

- **Scalability:** stateless app tier scales horizontally; queue-backed background work absorbs spikes; cost-tuned model routing (Opus for reasoning, Haiku for generation).
- **Reliability:** health-checked data tier; restart-on-failure services; graceful degradation (a missing model/render key surfaces a friendly message, never a crash).
- **Observability:** centralised logs/metrics; every agent run and content action recorded for full traceability.

Appendix A — Payload CMS v3 Security Posture

The platform's content backend is **Payload CMS v3 (3.85.0)**. Payload v3 is a secure, production-grade, actively-maintained headless CMS (Figma-backed) on the Next.js App Router; security is a shared responsibility — Payload provides the primitives, and this deployment configures and hardens them.

A.1 Built-in security controls (Payload v3)

Control	Detail
Authentication	bcrypt-hashed passwords · JWT in httpOnly cookies · API keys · email verification · password reset · account lockout (<code>maxLoginAttempts</code> / <code>lockTime</code>)
Authorization	Collection- and field-level access-control functions → real RBAC + ABAC
Injection defence	Parameterised queries via Drizzle ORM (Postgres)
Web security	CORS + CSRF controls, configurable origins; inherits the Next.js security model
Data integrity	Versioned drafts, editorial workflow state, audit hooks
Maintenance	Actively maintained; pinned, recent 3.85.0

A.2 Applied controls in this platform

Area	Configuration
Transport	HTTPS/TLS · <code>COOKIE_SECURE=true</code> · secure session cookies
Access control	RBAC (7 roles) + ABAC (site, department, locale) at collection and field level
Secrets	<code>PAYLOAD_SECRET</code> and provider keys held in environment, not committed to source
Governance	Editorial gating (draft → review → publish) · immutable audit log
Agentic safety	Agents propose drafts ; a human approves before publish
Sovereignty	Hosted in-Kingdom ; secrets isolated from content and code

A.3 Hardening checklist

- Keep Payload **patched** and monitor its security advisories.
- Strong, rotated `PAYLOAD_SECRET` ; restrict `/admin` exposure; MFA for maintainers.
- **Dependency / SCA scanning** (Dependabot / Snyk / `npm audit`) in CI.
- Replace the first-boot dev schema-push with **generated Payload migrations**.
- Edge **rate-limiting / WAF**; least-privilege database and registry credentials.

No framework is free of vulnerabilities; the correct posture is to **stay current and monitor advisories**. Most real-world risk lies in weak secrets, over-permissive access rules, leaked environment variables or unpatched dependencies — all addressed above.

Prepared for HUMAIN — Confidential. Master deck: Product Requirements, UI/UX & User Journeys, and High-Level Architecture for the HUMAIN Create Studio platform.